

NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health.



A Senior Project SDD for BSc. Degree in Software Engineering

**DEPARTMENT OF SOFTWARE ENGINEERING
COLLEGE OF COMPUTING AND INFORMATICS
HARAMAYA UNIVERSITY**

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HARAMAYA

DECLARATION SHEET

The **NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health**. SRS outlined herein is the collective effort of the group members listed below, completed without external assistance, under the supervision of **prof. Gemechis Teshite**.

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v. Definition, Abbreviation and Acronyms

1. Abbreviations and Acronyms

The following abbreviations are used throughout this document to refer to system components, roles, and technical standards:

- **AI:** Artificial Intelligence – Refers to the machine learning models used for sentiment analysis and trend detection.
- **API:** Application Programming Interface – The set of protocols used for communication between the mobile/web frontends and the backend services.
- **FR:** Functional Requirement – A specific behavior or function that the system must perform, as defined in the SRS.
- **JSONB:** JSON Binary – The data format used in PostgreSQL to store flexible, semi-structured metadata.
- **NFR:** Non-Functional Requirement – A requirement that specifies criteria that can be used to judge the operation of a system (e.g., security, performance).
- **PK / FK:** Primary Key / Foreign Key – Database markers used to uniquely identify records and establish relationships between tables.
- **RBAC:** Role-Based Access Control – A security approach that restricts system access to authorized users based on their role (Adolescent, Guardian, Counselor, or Admin).
- **SSL/TLS:** Secure Sockets Layer / Transport Layer Security – Encryption protocols for securing data in transit.
- **UML:** Unified Modeling Language – The standard notation used for the diagrams (Class, Sequence, Activity, etc.) in this document.
- **UUID:** Universally Unique Identifier – A 128-bit number used to uniquely identify information in the database.

2. Definitions

To ensure a common understanding among developers, advisors, and evaluators, the following terms are defined in the context of the NEURONET system:

- **Adolescent User:** The primary system user, aged 13–17, who uses the mobile application for journaling and mood tracking.
- **Counselor:** A mental health professional who monitors system-generated alerts and provides non-clinical guidance through secure channels.
- **Guardian:** A parent or legal caretaker responsible for managing the adolescent's account and granting/revoking data-sharing consent.
- **Sentiment Score:** A numerical value generated by the AI Analysis Subsystem representing the emotional tone of a journal entry.
- **System Boundary:** The conceptual line separating the internal NEURONET services from external entities like cloud infrastructure and third-party databases.
- **Traceability:** The ability to verify that every design element in the SDD is linked directly to a requirement in the SRS

CHAPTER ONE

INTRODUCTION

1.1 Document Scope

1.1.1 Overview of the Software Design Document

This Software Design Document (SDD) defines the architectural and detailed design of the **NEURONET – AI-Driven Early Detection and Support System for Adolescent Mental Health**. The document is derived directly from the approved Software Requirements Specification (SRS) and translates the functional and non-functional requirements into a structured technical design that guides system implementation.

The SDD serves as the primary technical reference for the NEURONET project, providing comprehensive documentation of system architecture, component design, data structures, interface specifications, and requirement traceability. It bridges the gap between requirements analysis and system implementation by explaining how each requirement will be realized through software design decisions.

This document ensures that all stakeholders—including developers, advisors, and evaluators—share a common understanding of the system’s technical structure. By documenting design decisions in advance, the SDD minimizes ambiguity, supports consistent development practices, and ensures that the final system faithfully satisfies all specified requirements.

1.1.2 Scope of Design Coverage

The scope of this Software Design Document encompasses the following design aspects of the NEURONET system:

- **System Architecture:** Overall architectural structure, design patterns, and layered organization of system components
- **Design Goals:** Translation of non-functional requirements into specific design objectives and strategies
- **Subsystem Decomposition:** Division of the system into cohesive subsystems with clearly defined responsibilities
- **4+1 Architectural Views:** Logical, Process, Development, Physical, and Use Case views
- **Data Design:** Logical and physical data models, database schemas, and data dictionary specifications
- **Human Interface Design:** User interface layouts, screen designs, and interaction patterns for all user roles
- **Requirement Traceability:** Mapping of functional requirements to system components

All design decisions strictly adhere to the non-clinical, ethical, and privacy-first constraints defined in the SRS. The system supports adolescent emotional awareness without providing medical diagnosis, therapy, or clinical intervention.

1.1.3 Relationship to the Software Requirements Specification

This SDD is directly derived from the NEURONET Software Requirements Specification and maintains strict traceability to the requirements defined therein. The relationship between the two documents follows established software engineering principles:

- The **SRS defines what the system must do**
- The **SDD defines how the system will be built**

Every architectural component, subsystem, and design element described in this document is traceable to one or more requirements from the SRS. No new requirements are introduced; instead, existing requirements are translated into implementable design specifications.

Key SRS elements informing this SDD include:

- Functional Requirements (FR-01 to FR-58)
- Non-Functional Requirements (NFR-01 to NFR-16)
- Use Case Model
- Data Requirements
- User Classes and Access Control Rules

Any future changes to system requirements must be reflected consistently in both documents to preserve traceability and academic integrity.

1.1.4 Out-of-Scope Design Elements

The following elements are intentionally excluded from this Software Design Document:

- Detailed source code and programming syntax
- Low-level algorithm optimization and performance tuning
- Deployment automation and CI/CD pipeline configurations
- Clinical decision-making or diagnostic logic
- Crisis intervention or emergency response mechanisms
- Integration with external healthcare or insurance systems

These exclusions ensure the design remains within the approved academic and non-clinical scope of the NEURONET project.

1.1.5 Recommended Reading Sequence

To ensure effective understanding, readers are encouraged to follow this reading sequence:

For All Readers

- Chapter One – Introduction
- Chapter Two, Section 2.1 – System Overview

For Developers

- Chapter Two, Sections 2.2–2.4
- Chapter Three – Data Design
- Chapter Four – Human Interface Design
- Chapter Five – Requirement Matrix

For Advisors and Evaluators

- Chapter Two, Section 2.2 – Design Goals
- Chapter Five – Requirement Traceability
- Other chapters as needed

For Future Maintainers

- Chapter Two – System Architecture
- Chapter Three – Data Structures
- Chapter Four – User Interaction Patterns

This sequence supports progressive understanding from high-level concepts to detailed design specifications.

1.2 Document Purpose

1.2.1 Purpose of the Software Design Document

The primary purpose of this Software Design Document is to provide a clear, structured, and technically sound blueprint for the development of the NEURONET system. It translates approved system requirements into detailed design specifications that guide implementation activities.

This document also serves as a communication medium that ensures developers, advisors, and evaluators share a common understanding of system architecture, component responsibilities, and design rationale.

1.2.2 Design Objectives

This SDD is prepared with the following objectives:

- **Modularity:** Separation of concerns through a modular architecture

- **Scalability:** Support for future growth in users and data volume
- **Security:** Integration of role-based access control and data protection
- **Ethical Compliance:** Preservation of adolescent privacy and autonomy
- **Maintainability:** Clear documentation to support long-term evolution
- **Traceability:** Explicit mapping between requirements and design elements

These objectives align directly with the system goals defined in the NEURONET SRS.

1.2.3 Support for Development and Implementation

For the development team, this document provides:

- Architectural structure and component boundaries
- Subsystem responsibilities and interface definitions
- Database schemas and data relationships
- User interface design guidance
- Integration and testing support

The SDD enables confident implementation while maintaining alignment with approved requirements and ethical constraints.

1.2.4 Support for Evaluation and Verification

The SDD supports academic evaluation by providing:

- Documented design rationale
- Clear requirement-to-design traceability
- Evidence of non-functional requirement fulfillment
- Verification of ethical and privacy compliance

This transparency enables objective assessment of the system design.

1.2.5 Role within the Project Documentation Set

Within the NEURONET project documentation:

- The **SRS** defines requirements
- The **SDD** defines design
- Implementation artifacts realize the design
- Test documents verify correctness

The SDD serves as the authoritative design reference and foundation for all implementation activities.

1.3 Document Conventions

This section describes the standards, formatting rules, and notation conventions followed throughout this Software Design Document. These conventions are applied consistently to improve readability, ensure clarity, and maintain alignment with academic software engineering documentation standards.

1.3.1 Formatting Conventions

The Software Design Document follows standardized formatting conventions to ensure professionalism and consistency.

Document Structure

- The document is organized into clearly defined chapters and sections following the approved SDD outline
- Hierarchical numbering is used for sections and subsections (e.g., 1.3, 1.3.1, 2.4.2)
- Each chapter begins with a logical progression from overview to detailed design

Text Formatting

- The document uses formal academic English suitable for technical documentation
- Paragraphs are used for explanations, while bullet points are used for structured listings
- Headings and subheadings are consistently styled to reflect document hierarchy
- Emphasis is achieved through context rather than excessive formatting

Tables and Figures

- Tables and figures are numbered sequentially within each chapter
- Each table and figure includes a descriptive title
- All visual elements are referenced within the text before their appearance
- Tables are used for structured data such as mappings and specifications

1.3.2 Terminology and Naming Conventions

Consistent terminology and naming conventions are used throughout the document to avoid ambiguity and misinterpretation.

System Terminology

- **System** refers to the complete NEURONET platform
- **Subsystem** refers to a major functional division within the system
- **Component** refers to a specific module or service
- **Module** and **Service** are used interchangeably for backend elements

User Role Terminology

- **Adolescent User:** Primary system user aged 13–17
- **Guardian:** Parent or legal caretaker responsible for consent and oversight
- **Counselor:** Mental health professional providing guidance and monitoring
- **System Administrator:** User responsible for technical management and configuration

Naming Standards

- Entity and class names use **PascalCase** (e.g., UserProfile, JournalEntry)
- Attribute names use **PascalCase** for consistency (e.g., UserID, CreatedAt)
- Database table names match entity names to support traceability
- Acronyms are written in uppercase and defined upon first use

1.3.3 UML and Diagram Conventions

Unified Modeling Language (UML) is used as the standard notation for representing system structure and behavior.

Diagram Types Used

- Use Case Diagrams – Functional interaction from user perspective
- Class Diagrams – Static structure and relationships
- Sequence Diagrams – Interactions between components over time
- Activity Diagrams – Workflow and process logic
- State Diagrams – Object lifecycle behavior
- Component Diagrams – Software module organization
- Deployment Diagrams – Physical system architecture
- Package Diagrams – Subsystem grouping and dependencies

Notation Standards

- All diagrams follow UML 2.0 standards
- Actors are represented using stick figures
- Classes are represented using rectangles with compartments
- Relationships use standard UML arrows and connectors

Each diagram includes a title and figure number and is referenced within the corresponding chapter.

1.3.4 Requirement and Design Traceability Conventions

To ensure traceability between requirements and design elements, consistent referencing conventions are applied.

Requirement Identification

- Functional Requirements are referenced as **FR-XX**
- Non-Functional Requirements are referenced as **NFR-XX**
- Use Cases are referenced as **UC-XX**

Traceability Rules

- Design components reference related requirements where applicable
- The Requirement Matrix in Chapter Five provides complete mappings
- Traceability is maintained bidirectionally (requirement-to-design and design-to-requirement)

These conventions ensure transparency and verifiability of design decisions.

1.3.5 Document Conventions Summary

Table 1 Document Conventions Summary

Convention Type	Standard Applied
Section Numbering	Hierarchical decimal notation
Figure Numbering	Chapter-based sequential numbering
Table Numbering	Chapter-based sequential numbering
Entity Naming	PascalCase
Requirement References	FR-XX, NFR-XX
UML Notation	UML 2.0 standard
Language Style	Formal academic English

1.4 Intended Audience

This Software Design Document is intended for multiple stakeholders involved in the development, evaluation, and future maintenance of the NEURONET system. Each section of the document is structured to address the specific information needs of these audiences while maintaining overall technical consistency and clarity.

1.4.1 Primary Audience: Development Team

The primary audience for this Software Design Document is the project development team responsible for implementing the NEURONET system.

For developers, this document provides:

- **Architectural Guidance:** A clear understanding of the system architecture, component boundaries, and integration patterns required for implementation.
- **Component Specifications:** Detailed descriptions of subsystem responsibilities, interfaces, and dependencies that guide modular development.

- **Data Structure Definitions:** Complete logical and physical data models that support accurate database implementation.
- **Interface Requirements:** User interface specifications that ensure consistency across mobile and web platforms.
- **Integration Context:** Insight into how individual components interact within the overall system.

Developers should use this document as the primary technical reference throughout the implementation phase to ensure alignment with approved design decisions and requirements.

1.4.2 Academic Advisor

The academic advisor serves as a key reviewer and supervisor of the NEURONET project.

For the advisor, this document provides:

- **Design Visibility:** Clear documentation of architectural and design decisions, including the rationale behind them.
- **Feasibility Assessment:** Technical details that support evaluation of the system's feasibility within academic and time constraints.
- **Quality Evaluation:** Evidence of adherence to accepted software engineering principles and documentation standards.
- **Progress Monitoring:** A baseline design reference against which implementation progress can be evaluated.

The advisor may use this document to provide guidance, feedback, and validation throughout the project lifecycle.

1.4.3 Examiners and Evaluators

Academic examiners and evaluation committee members represent a critical audience for this document.

For evaluators, the SDD enables:

- **Design Quality Assessment:** Evaluation of architectural soundness, modularity, and system organization.
- **Requirement Verification:** Confirmation that all SRS requirements are adequately addressed through design elements.
- **Constraint Compliance:** Verification that ethical, privacy, and non-clinical constraints are respected.
- **Traceability Validation:** Assessment of the accuracy and completeness of requirement-to-design mappings.
- **Technical Rigor Evaluation:** Judgment of the depth, clarity, and professionalism of the design documentation.

The structured presentation of design details supports objective and transparent academic evaluation.

1.4.4 Future Developers and Maintainers

Although developed as an academic project, the NEURONET system may be extended or maintained in the future.

For future developers and maintainers, this document provides:

- **System Understanding:** Comprehensive documentation of system architecture and design rationale that supports knowledge transfer.
- **Modification Guidance:** Clearly defined component boundaries and interfaces that indicate where changes can be safely applied.
- **Extension Points:** Identification of areas where functionality can be enhanced without affecting existing features.
- **Maintenance Support:** Data models and interface specifications that assist with debugging and system maintenance.

This ensures the long-term usefulness of the document beyond the initial project implementation.

1.4.5 Summary of Audience Considerations

Each section of this Software Design Document is written to balance technical depth with clarity, ensuring accessibility to readers with different technical backgrounds and responsibilities. The document structure allows selective reading, enabling each audience to focus on the sections most relevant to their role while maintaining access to the complete system design.

By addressing the needs of developers, advisors, evaluators, and future maintainers, this document serves as a comprehensive and reliable design reference for the NEURONET project throughout its lifecycle.

CHAPTER TWO

SYSTEM ARCHITECTURAL DESIGN

2.1 System Overview

2.1.1 Introduction

The NEURONET system is an AI-assisted digital platform designed to support adolescent emotional awareness through secure journaling, controlled communication, and intelligent pattern detection. From a design perspective, the system is architected to provide a modular, scalable, and ethically responsible solution that aligns with non-clinical mental health support objectives.

The system is implemented as a multi-tier architecture consisting of client applications, backend services, AI processing components, and data storage layers. This architectural separation ensures maintainability, security, and role-based control while supporting multiple user types with distinct permissions and responsibilities.

NEURONET operates on the principle of ethical, non-intrusive support. The system provides adolescents with tools for self-expression and reflection while enabling guardians and counselors to provide appropriate oversight and support within carefully defined boundaries. All AI functionality is limited to pattern detection and informational responses, explicitly excluding diagnosis, therapy, or clinical advice.

2.1.2 System Objectives from a Design Perspective

From a software design standpoint, NEURONET aims to achieve the following objectives:

- **Secure Role-Based Interactions:** Support distinct interaction patterns for adolescents, guardians, counselors, and administrators with appropriate access controls and data visibility.
- **Private Self-Expression:** Enable adolescents to engage in journaling and mood reflection through secure, private interfaces that protect autonomy and confidentiality.
- **Ethical AI Integration:** Integrate AI-driven sentiment and trend analysis without performing clinical diagnosis or providing therapeutic advice.
- **Consent-Driven Communication:** Provide controlled communication channels that respect consent requirements, privacy constraints, and ethical boundaries.
- **Efficient Professional Support:** Allow counselors to manage multiple adolescents efficiently through structured assignments and dashboards.
- **Robust Data Protection:** Maintain high standards of data security, encryption, and access control to protect sensitive adolescent information.

These objectives directly influence architectural decisions, subsystem boundaries, technology selection, and implementation strategies.

2.1.3 High-Level System Architecture

The NEURONET system follows a layered architecture pattern, organizing components into distinct tiers with well-defined responsibilities and interfaces.

Figure 2.1: NEURONET High-Level System Architecture

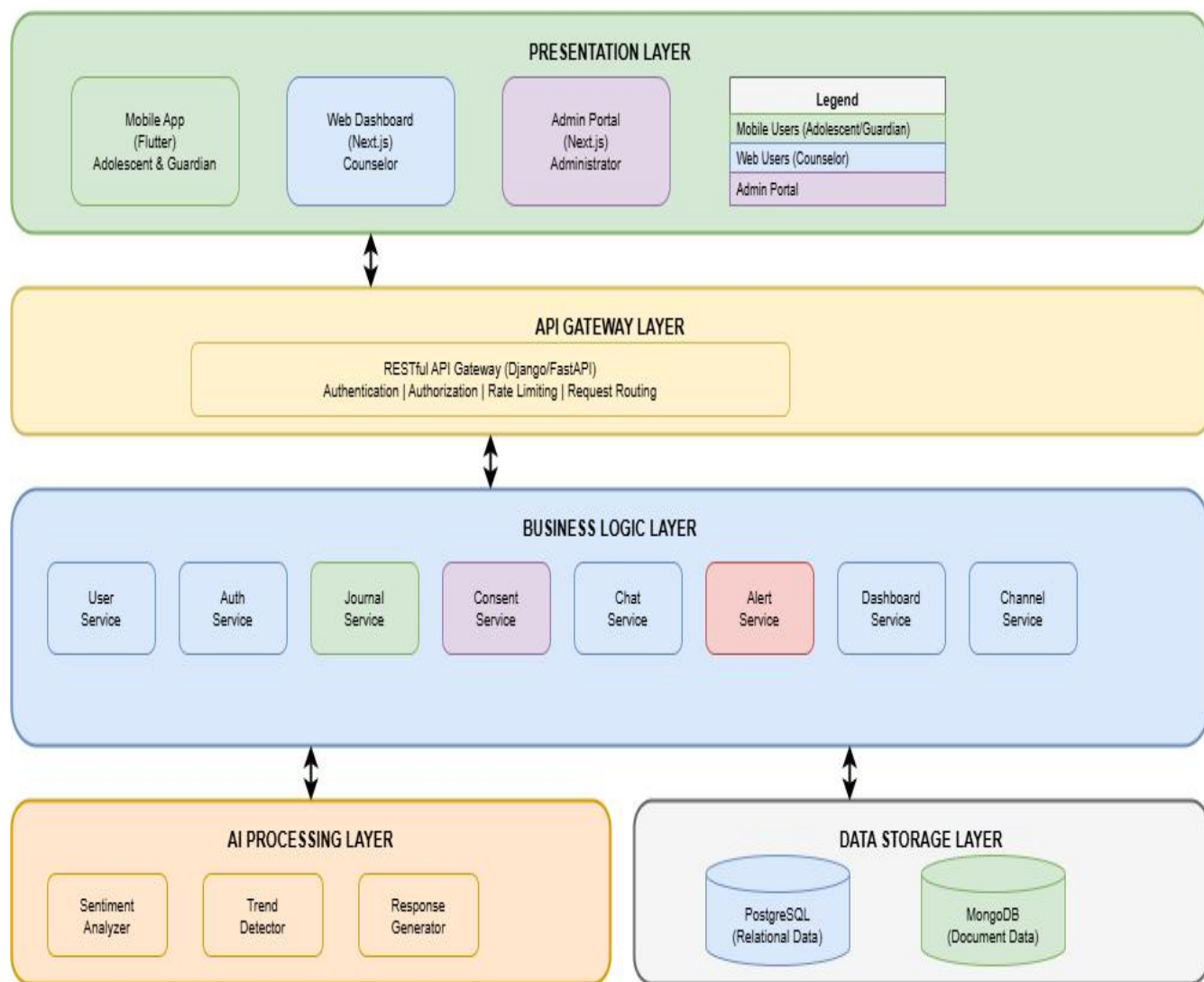


Figure 1 High-Level System Architecture Diagram

Layer Descriptions

1. Presentation Layer

- Mobile application for adolescent users and guardians
- Web-based dashboards for counselors and administrators
- Handles user interaction, input validation, and visualization
- Enforces role-specific interfaces and permissions

2. Application Layer

- RESTful API gateway handling all client requests
- Core business logic services (user, journal, chat, consent, alerts)
- Role-based access control and workflow coordination

3. AI Processing Layer

- Sentiment analysis of journal entries and messages
- Trend detection for emotional patterns
- Informational response generation
- Operates asynchronously and without direct user control

4. Data Layer

- Relational database for structured system data
- NoSQL database for unstructured journal and AI analysis data
- Ensures integrity, confidentiality, and auditability

2.1.4 User Roles and System Interaction Overview

2.1.4.1 External Actors

The NEURONET system interacts with four primary categories of external users. Each user category accesses the system through specific interfaces and is granted permissions based on clearly defined roles.

Adolescent Users (Ages 13–17)

- Primary system users who engage in journaling and mood-tracking activities
- Access the system exclusively through the Android mobile application
- Submit journal entries and record daily mood reflections
- Interact with the AI chat interface for informational support
- View personal dashboards showing trends and insights
- Communicate with assigned counselors through secure messaging
- Maintain full control over private content through consent-based sharing mechanisms

Guardian Users (Parents or Legal Caretakers)

- Responsible adults who register and manage adolescent accounts
- Access the system through the Android mobile application
- Grant or revoke consent for counselor access to adolescent data
- Receive alert notifications when concerning behavioral patterns are detected
- View aggregated summaries and trends without access to raw journal content
- Are restricted from accessing private adolescent data without explicit consent

Counselor Users (Mental Health Professionals)

- Licensed professionals providing non-clinical guidance and support
- Access the system through a web-based dashboard on desktop or laptop devices
- View adolescent data only when guardian consent has been granted
- Create educational channels and publish supportive informational content
- Communicate with adolescents and guardians via secure messaging
- Receive, review, and respond to system-generated alerts

System Administrators

- Technical personnel responsible for system management and oversight
- Access the system through an administrative web portal on desktop or laptop devices
- Manage user accounts, system configurations, and security policies
- Monitor system health, performance metrics, and audit logs
- Are strictly restricted from accessing private user content such as journals, messages, or mood records

2.1.5 Overall System Workflow

The high-level operational workflow includes:

1. Guardian registers adolescent and sets consent
2. Adolescent activates account
3. Adolescent records journals and moods
4. AI analyzes data asynchronously
5. Alerts generated based on trends
6. Counselors review insights and respond
7. Guardians receive summarized updates
8. System logs and enforces policies continuously

2.1.6 System Context Diagram

The System Context Diagram illustrates the NEURONET system as a single unified entity and depicts its interactions with external actors and systems. This high-level representation establishes the system boundary and identifies all external entities that exchange information with the NEURONET system.

Figure 2.2: NEURONET System Context Diagram

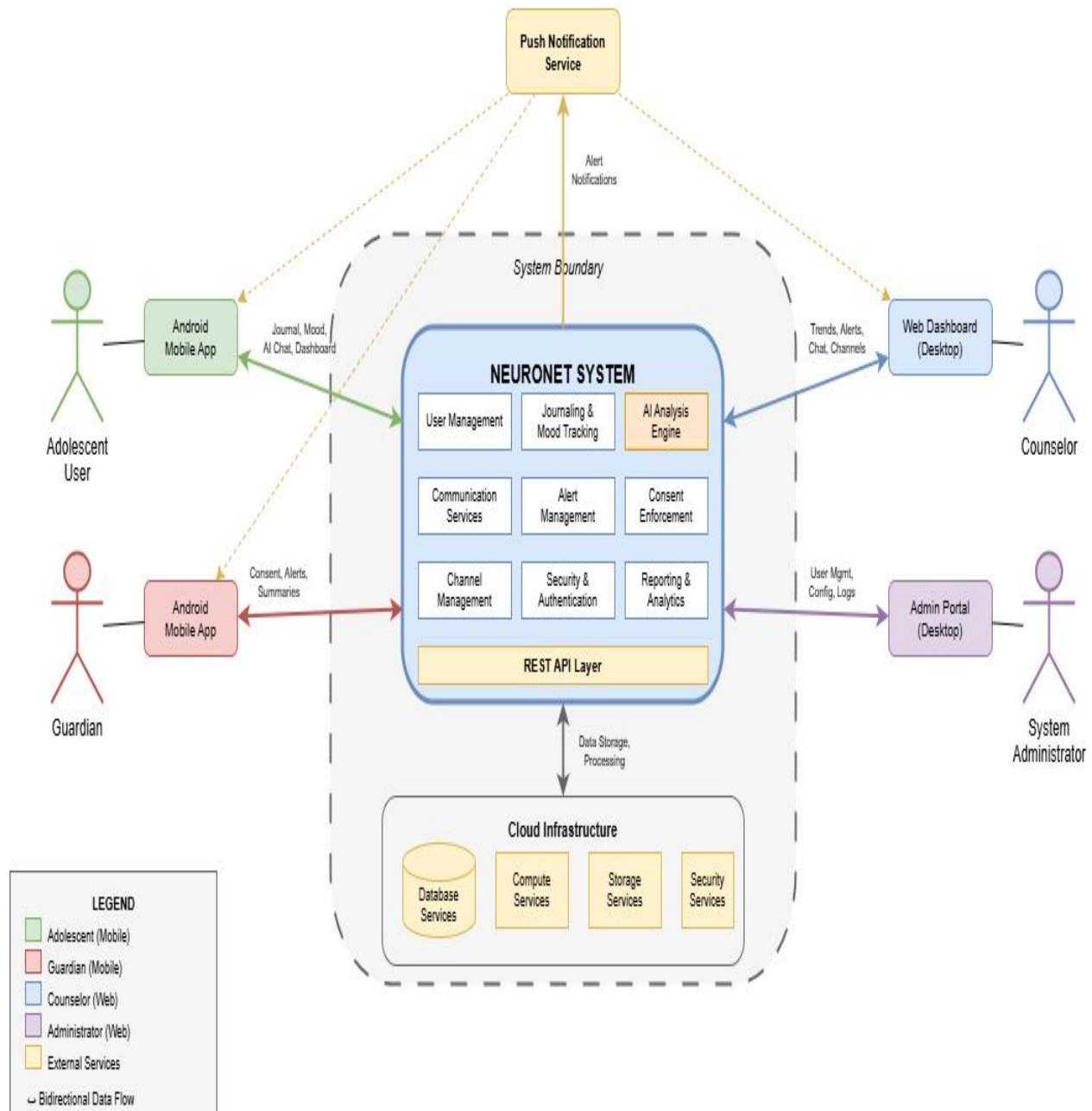


Figure 2 System Context Diagram

2.1.6.1 System Boundary

The NEURONET system boundary includes all internal components responsible for processing, storing, analyzing, and managing adolescent mental health support functionality.

Core System Services include:

- **User Management:** Registration, authentication, profile management, and role-based access control
- **Journaling and Mood Tracking:** Secure collection, storage, and retrieval of adolescent emotional data
- **AI Analysis and Insights:** Natural language processing and sentiment analysis of journal entries and interactions
- **Communication Services:** Secure messaging between adolescents, counselors, and guardians
- **Alert Management:** Generation, routing, and tracking of concern-based notifications
- **Consent Enforcement:** Guardian-controlled permissions governing data visibility and access
- **Channel Management:** Counselor-created educational content and community interaction features

2.1.6.2 External Systems and Infrastructure

Cloud Infrastructure

The NEURONET system relies on cloud-based infrastructure services to ensure availability, scalability, and security.

Key infrastructure components include:

- **Compute Services:** Application hosting and processing resources
- **Database Services:** PostgreSQL for structured relational data and MongoDB for unstructured content
- **Security Services:** SSL/TLS encryption, firewall protection, and access management mechanisms
- **Storage Services:** Secure file storage for media assets, backups, and system artifacts

2.1.6.3 Data Flow Summary

The following data flows occur across the system boundary between external actors and the NEURONET system.

Inbound Data Flows:

- User credentials and authentication requests
- Journal entries and mood records submitted by adolescents
- Consent decisions submitted by guardians
- Chat messages from adolescents, counselors, and guardians
- Administrative commands and system configuration inputs

Outbound Data Flows:

- Personalized dashboards and summarized insights delivered to users
- Alert notifications sent to guardians and counselors
- AI-generated responses and analysis results
- System reports, logs, and audit records provided to administrators

2.1.7 Design Considerations and Constraints

Ethical Constraints

- No diagnosis or therapy
- Privacy preserved even from guardians
- Explicit consent required for data sharing

Privacy Requirements

- Role-based access control
- Encryption in transit and at rest
- Comprehensive audit logging

Academic Constraints

- MVP-focused scope
- Limited development timeline
- Use of academic-appropriate technologies

Technical Constraints

- Moderate scalability targets
- Pre-trained non-clinical AI models
- Deployment on standard infrastructure

2.1.8 Summary of System Overview

The NEURONET system architecture balances ethical responsibility, modularity, and technical robustness. The layered design supports scalability and maintainability while enforcing strict privacy and non-clinical boundaries, providing a solid foundation for implementation and future enhancement.

2.2 Design Goals

2.2.1 Overview of Design Goals

The design goals of the NEURONET system are derived directly from the non-functional requirements specified in the Software Requirements Specification (SRS). These goals guide architectural decisions, technology selection, subsystem organization, and implementation strategies to ensure that the system operates efficiently, securely, and ethically while meeting user and academic expectations.

Given the sensitive nature of adolescent emotional and behavioral data, the system design places strong emphasis on privacy, security, usability, and ethical responsibility. Each design goal targets a specific quality attribute and collectively contributes to the overall robustness, reliability, and trustworthiness of the system.

Table 2.1: Design Goals and Achievement Strategies

Design Goal	Related NFRs	Achievement Strategy
Usability	NFR-06 – NFR-10	User-centered design, intuitive navigation, supportive language
Security	NFR-11 – NFR-16	Role-based access control, encryption, secure authentication
Performance	NFR-01 – NFR-05	Optimized APIs, asynchronous processing, caching
Scalability	NFR-05	Modular architecture, stateless services
Reliability	Quality Attribute	Error handling, backup and recovery mechanisms
Maintainability	Quality Attribute	Modular design, documentation, coding standards
Ethical Compliance	BR-01 – BR-19	Non-clinical constraints, consent enforcement, privacy-first design

2.2.2 Usability and User-Centered Design Goal

One of the primary design goals of NEURONET is to ensure high usability for all user classes, particularly adolescent users who may have varying levels of technical literacy and emotional readiness.

From a design perspective, usability is addressed through simplicity, clarity, and consistency across all user interfaces.

Design Strategies:

Interface Simplicity

- Mobile interfaces are designed with minimal interaction steps
- Clear visual hierarchy guides users through common tasks
- Consistent navigation patterns are used across all screens

Role-Appropriate Content

- Each user role accesses only features relevant to their responsibilities
- Information is presented at appropriate complexity levels
- Technical terminology is avoided in user-facing interfaces

Supportive Communication

- All system messages use non-judgmental and supportive language
- Alerts and insights are presented calmly and informatively
- Error messages provide constructive guidance for resolution

Accessibility Considerations

- Readable font sizes and appropriate color contrast
- Touch-friendly interface components
- Logical navigation order and accessibility support

This user-centered design approach reduces cognitive load and encourages sustained engagement with the system.

2.2.3 Security and Privacy Design Goal

Security and privacy are central pillars of the NEURONET system design. The system is architected to protect sensitive adolescent data through multiple layers of technical and procedural safeguards.

Design Strategies:

Role-Based Access Control (RBAC)

- Clearly defined permissions for each user role
- Strict enforcement of data visibility rules
- Guardians cannot access raw adolescent journal entries
- Administrators have no access to private user content

Data Encryption

- Sensitive data encrypted at rest using industry-standard algorithms
- All network communication secured using HTTPS/TLS

- Passwords securely hashed using modern cryptographic techniques

Authentication and Session Management

- Secure authentication mechanisms using token-based sessions
- Automatic session expiration after inactivity
- Protection against unauthorized access and session hijacking

Audit and Monitoring

- Logging of all critical system actions
- Audit logs protected from unauthorized modification
- Monitoring mechanisms to detect suspicious behavior

These measures ensure confidentiality, integrity, and accountability throughout the system.

2.2.4 Ethical AI Usage Design Goal

NEURONET integrates artificial intelligence components while strictly adhering to ethical and non-clinical constraints. AI is used only as a supportive analytical tool and never as a decision-making authority.

Design Strategies:

Limited AI Scope

- AI outputs restricted to sentiment indicators and trend summaries
- No diagnosis, therapy, or medical recommendations
- No evaluative or judgmental conclusions about user behavior

Clear Distinction Between AI and Human Judgment

- AI-generated insights are clearly labeled
- Counselors' decisions are explicitly human-driven
- Users are informed about AI limitations

Safeguard Mechanisms

- AI redirects clinical or emergency queries to human support
- Predefined responses for sensitive topics
- Continuous review of AI outputs for appropriateness

Transparency

- Users are informed about what data AI processes
- Clear explanations of how insights are generated

- No hidden or opaque AI decision-making processes

This design ensures responsible AI use while maintaining user trust and ethical integrity.

2.2.5 Performance and Responsiveness Design Goal

The system is designed to provide timely responses and maintain stable performance under expected workloads.

Design Strategies:

Efficient API Design

- RESTful APIs optimized for common operations
- Pagination used for large datasets
- Lightweight responses to reduce network overhead

Asynchronous Processing

- AI analysis tasks handled in the background
- Non-critical tasks processed asynchronously
- Queue-based job handling for scalability

Database Optimization

- Indexed queries for frequently accessed data
- Connection pooling to reduce overhead
- Optimized data retrieval for dashboards and reports

Caching Strategy

- Frequently accessed data cached where appropriate
- Cache invalidation strategies to maintain consistency
- Reduced load on primary databases

Performance Targets

- Standard API response time under two seconds
- Dashboard loading within acceptable thresholds
- AI analysis completed within reasonable timeframes

2.2.6 Scalability and Maintainability Design Goal

The NEURONET architecture is designed to support future growth in users, data volume, and functionality without requiring fundamental redesign.

Design Strategies:

Modular Architecture

- Clear separation of subsystems and responsibilities
- Well-defined interfaces between components
- Independent development and testing of modules

Stateless Services

- Application services designed to be stateless
- Session state managed externally
- Supports horizontal scaling and load balancing

Technology Standards

- Use of widely supported, open-source technologies
- Frameworks with strong community support
- Standardized APIs and protocols

Documentation and Code Quality

- Comprehensive design documentation
- Clear coding standards and conventions
- Well-documented interfaces and services

2.2.7 Reliability and Availability Design Goal

Reliability is essential to maintain user confidence and continuous system operation.

Design Strategies:

Error Handling

- Robust exception handling across all components
- Graceful degradation in case of partial failures
- User-friendly error reporting

Data Protection

- Regular automated database backups
- Support for data recovery and rollback
- Validation to prevent data corruption

Monitoring and Alerting

- Continuous monitoring of system health
- Performance and error metrics collection
- Automated alerts for abnormal conditions

Recovery Mechanisms

- Automatic service restart upon failure
- Support for database failover strategies
- Transaction management for data consistency

2.2.8 Compliance and Academic Constraints Design Goal

As a final-year academic project, NEURONET must adhere to institutional, ethical, and feasibility constraints.

Design Strategies:

Academic Standards

- Documentation aligned with departmental guidelines
- Clearly justified and traceable design decisions
- Proper separation of requirements and design

Technology Constraints

- Use of free and open-source tools
- Avoidance of proprietary dependencies
- Technologies suitable for academic evaluation

Ethical Research Guidelines

- Strong privacy-preserving design
- Explicit consent mechanisms
- Protection of adolescent autonomy

Feasibility

- Scope limited to a realistic MVP
- Design aligned with available time and resources
- Clear boundaries on system capabilities

2.2.9 Summary of Design Goals

The design goals of the NEURONET system collectively ensure that the system is usable, secure, scalable, reliable, and ethically responsible. By aligning architectural decisions with non-functional requirements and ethical constraints, these goals provide a strong foundation for effective implementation and future system evolution while maintaining trust among all stakeholders.

2.3 Subsystem Decomposition

2.3.1 Overview of Subsystem Decomposition

The NEURONET system is decomposed into multiple subsystems to manage system complexity, improve maintainability, and support independent development and testing. Each subsystem encapsulates a cohesive set of responsibilities related to a specific functional or architectural concern within the system.

This modular decomposition approach provides the following benefits:

- Clear separation of concerns between functional domains
- Independent development, testing, and maintenance of subsystems
- Improved scalability and extensibility
- Simplified enforcement of role-based access control
- Isolation of sensitive data handling and ethical constraints

Subsystem decomposition ensures that each major feature of the NEURONET system can evolve independently while maintaining well-defined interfaces with other subsystems.

2.3.2 Identified Subsystems

Based on the functional and non-functional requirements specified in the SRS, the NEURONET system is divided into the following major subsystems:

1. User Management Subsystem
2. Authentication and Authorization Subsystem
3. Consent Management Subsystem
4. Journaling and Mood Tracking Subsystem
5. Communication and Chat Subsystem
6. AI Analysis and Insight Subsystem
7. Alert and Notification Subsystem
8. Counselor Assignment and Channel Subsystem
9. Dashboard and Visualization Subsystem
10. Administration and System Monitoring Subsystem

Each subsystem is designed with a clearly defined responsibility and interacts with other subsystems through controlled interfaces.

Table 2 Subsystem Responsibilities Summary

Subsystem	Primary Responsibilities
User Management	User profiles, roles, account lifecycle
Authentication and Authorization	Login, logout, RBAC, session management
Consent Management	Consent recording, enforcement, audit
Journaling and Mood Tracking	Journal entries, mood records, history
Communication and Chat	AI chat, counselor chat, messaging
AI Analysis and Insight	Sentiment analysis, trend detection, insights
Alert and Notification	Alert generation, delivery, history
Counselor Assignment and Channel	Assignments, channels, group interactions
Dashboard and Visualization	Role-based dashboards, data visualization
Administration and Monitoring	User administration, system configuration, logs

2.3.3 User Management Subsystem

The User Management Subsystem is responsible for maintaining core user information for all system roles, including adolescents, guardians, counselors, and administrators.

Responsibilities:

- Manage user profile information
- Maintain user roles and account status
- Support profile updates within permission limits
- Associate users with system activities and data
- Manage user lifecycle events such as creation, activation, and deactivation

Related Requirements: FR-01, FR-02, FR-09, FR-10, FR-11, FR-12

Interfaces:

- Provides user data to all other subsystems
- Receives authentication status from the Authentication subsystem
- Integrates with Consent Management for guardian–adolescent relationships

2.3.4 Authentication and Authorization Subsystem

This subsystem ensures secure access to the system and enforces role-based permissions across all operations.

Responsibilities:

- Authenticate users during login
- Verify credentials securely
- Generate and validate authentication tokens
- Enforce role-based access control (RBAC)
- Manage user sessions and timeouts
- Handle logout and session termination

Related Requirements: FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-56, FR-57

Interfaces:

- Integrates with User Management for credential validation
- Provides authorization context to all subsystems
- Logs authentication events to the Administration subsystem

2.3.5 Consent Management Subsystem

The Consent Management Subsystem governs ethical data usage and communication permissions within the system.

Responsibilities:

- Record guardian consent for data sharing
- Record adolescent consent preferences
- Enforce consent rules before data access
- Support consent revocation at any time
- Maintain consent history for auditing
- Validate consent status for protected operations

Related Requirements: FR-16, FR-17, FR-18, FR-19, FR-20, BR-05, BR-06, BR-07, BR-08

Interfaces:

- Integrates with User Management for guardian relationships
- Provides consent validation to Communication, Alert, and Dashboard subsystems
- Logs consent changes to Administration subsystem

2.3.6 Journaling and Mood Tracking Subsystem

This subsystem enables adolescents to engage in private self-expression and emotional reflection.

Responsibilities:

- Create and store journal entries securely
- Record mood reflections with timestamps
- Retrieve personal journal and mood history
- Prepare textual data for AI analysis
- Enforce strict privacy on raw content

Related Requirements: FR-21, FR-22, FR-23, FR-24, FR-25

Interfaces:

- Receives user context from Authentication subsystem
- Provides data to AI Analysis subsystem
- Integrates with Dashboard subsystem for visualization

2.3.7 Communication and Chat Subsystem

The Communication and Chat Subsystem manages all text-based interactions in the system.

Responsibilities:

- Support adolescent-to-AI informational chat
- Manage adolescent-to-counselor private chat (with consent)
- Support guardian-to-counselor communication
- Store and retrieve message history
- Enforce consent and approval rules

Related Requirements: FR-26, FR-27, FR-28, FR-29, FR-30, FR-34, FR-37, FR-38

Interfaces:

- Integrates with AI Analysis subsystem
- Validates permissions via Consent Management
- Provides chat data to Dashboard subsystem

2.3.8 AI Analysis and Insight Subsystem

This subsystem performs non-clinical analysis on user-generated textual data.

Responsibilities:

- Perform sentiment analysis on journal entries
- Analyze chat messages for emotional indicators
- Detect behavioral trends over time
- Generate non-diagnostic summaries
- Provide informational AI responses
- Redirect clinical or emergency queries

Related Requirements: FR-26–FR-33, BR-13–BR-16

Interfaces:

- Receives data from Journaling and Communication subsystems
- Provides insights to Dashboard and Alert subsystems

2.3.9 Alert and Notification Subsystem

This subsystem supports early awareness through controlled alert generation.

Responsibilities:

- Monitor AI analysis outputs
- Generate alerts based on thresholds
- Route alerts according to consent rules
- Deliver notifications to authorized users
- Maintain alert history

Related Requirements: FR-48, FR-49, FR-50, FR-51

Interfaces:

- Receives triggers from AI Analysis
- Validates consent before alert delivery
- Logs alert activities to Administration subsystem

2.3.10 Counselor Assignment and Channel Subsystem

This subsystem manages professional support structures.

Responsibilities:

- Assign counselors to adolescents
- Manage counselor workload
- Create and manage informational channels
- Support adolescent participation and interaction

Related Requirements: FR-35, FR-36, FR-39, FR-40, FR-41, FR-42

Interfaces:

- Integrates with User Management
- Provides data to Dashboard subsystem
- Supports Communication subsystem

2.3.11 Dashboard and Visualization Subsystem

This subsystem presents system insights in a role-appropriate manner.

Responsibilities:

- Display adolescent personal dashboards
- Provide aggregated guardian views
- Present counselor trend summaries
- Visualize alerts and patterns
- Enforce role-based data filtering

Related Requirements: FR-43, FR-44, FR-45, FR-46, FR-47

2.3.12 Administration and System Monitoring Subsystem

This subsystem supports system oversight and maintenance.

Responsibilities:

- Manage user accounts and roles
- Configure counselor assignments
- Monitor system performance

- Manage system logs
- Restrict admin access to private data

Related Requirements: FR-52, FR-53, FR-54, FR-55, FR-58

2.3.13 Subsystem Package Diagram

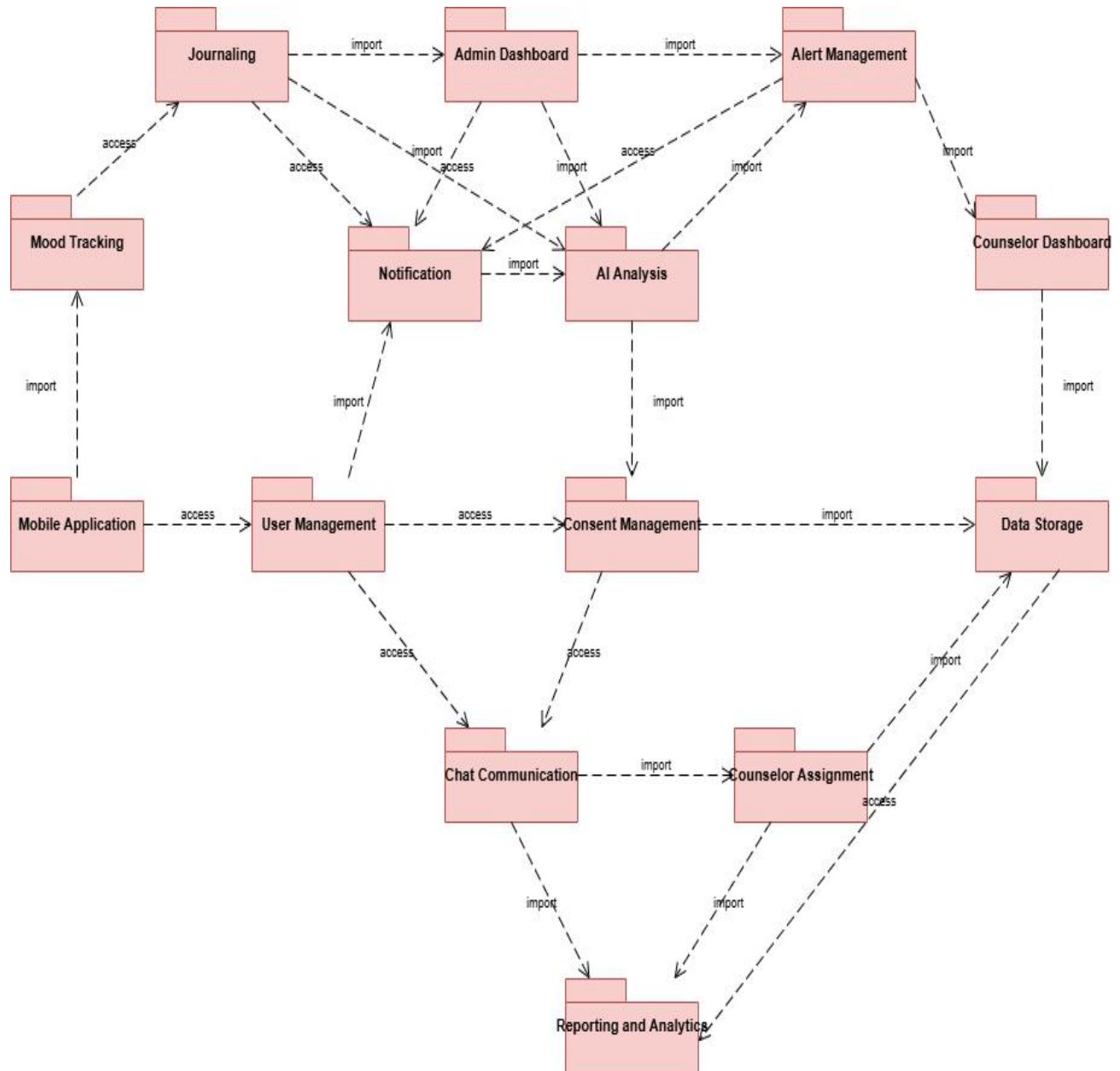


Figure 2.3 Package Diagram for NEURONET System

Figure 3 Subsystem Package Diagram

2.3.14 Subsystem Interaction Summary

Subsystems interact through clearly defined interfaces to minimize coupling. AI Analysis consumes data from Journaling and Communication subsystems, Alerts depend on AI outputs and consent validation, and Dashboards aggregate filtered data across subsystems. Authentication and Consent enforcement are applied consistently across all protected operations.

2.3.15 Summary of Subsystem Decomposition

The subsystem decomposition of NEURONET establishes a scalable, modular, and ethically responsible system architecture. By clearly separating responsibilities and enforcing controlled interactions, this design supports maintainability, security, and future system evolution. This structure provides the foundation for the detailed architectural views presented in Section 2.4.

2.4 4+1 Architecture View Model

The NEURONET system architecture is documented using the **4+1 Architecture View Model**, a widely adopted architectural framework that presents the system from multiple complementary perspectives. This model ensures clarity, completeness, and alignment with stakeholder concerns, including users, developers, system designers, and academic evaluators.

The “+1” **Use Case View** captures system functionality from the perspective of external actors and serves as the foundation for the remaining four views: **Logical View**, **Process View**, **Development View**, and **Physical View**. Each view addresses a specific architectural concern and is represented using appropriate UML artifacts.

2.4.1 Logical View

The Logical View describes the **static structure** of the NEURONET system. It models the core entities, their attributes, relationships, and constraints, independent of implementation or deployment considerations. This view ensures that system functionality is supported by a coherent internal data and object model.

2.4.2.1 Class Diagram – Core System Entities

The class diagram represents the primary domain entities of the system, including users, journals, AI analysis results, alerts, consents, and communication artifacts. It also illustrates inheritance relationships among user roles and associations between functional components.

NEURONET - Class Diagram (Core System Entities)

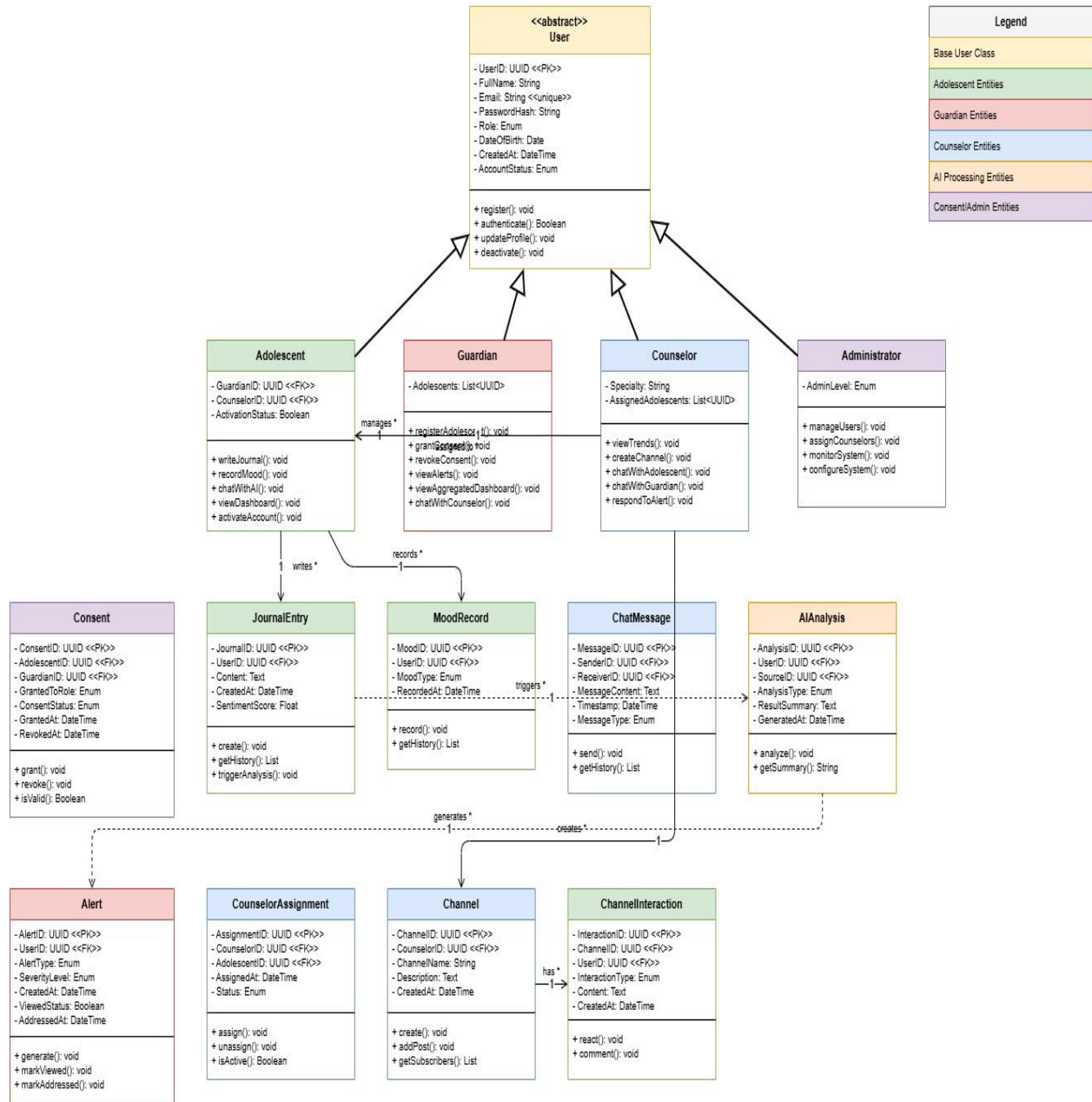


Figure 4 Class Diagram – Core System Entities

2.4.2.2 State Diagram – Consent Lifecycle

This state diagram models the lifecycle of consent, including granting, validation, revocation, and expiration. It ensures that all data access and communication actions are ethically governed.

Figure 2.10: State Diagram - Consent Lifecycle

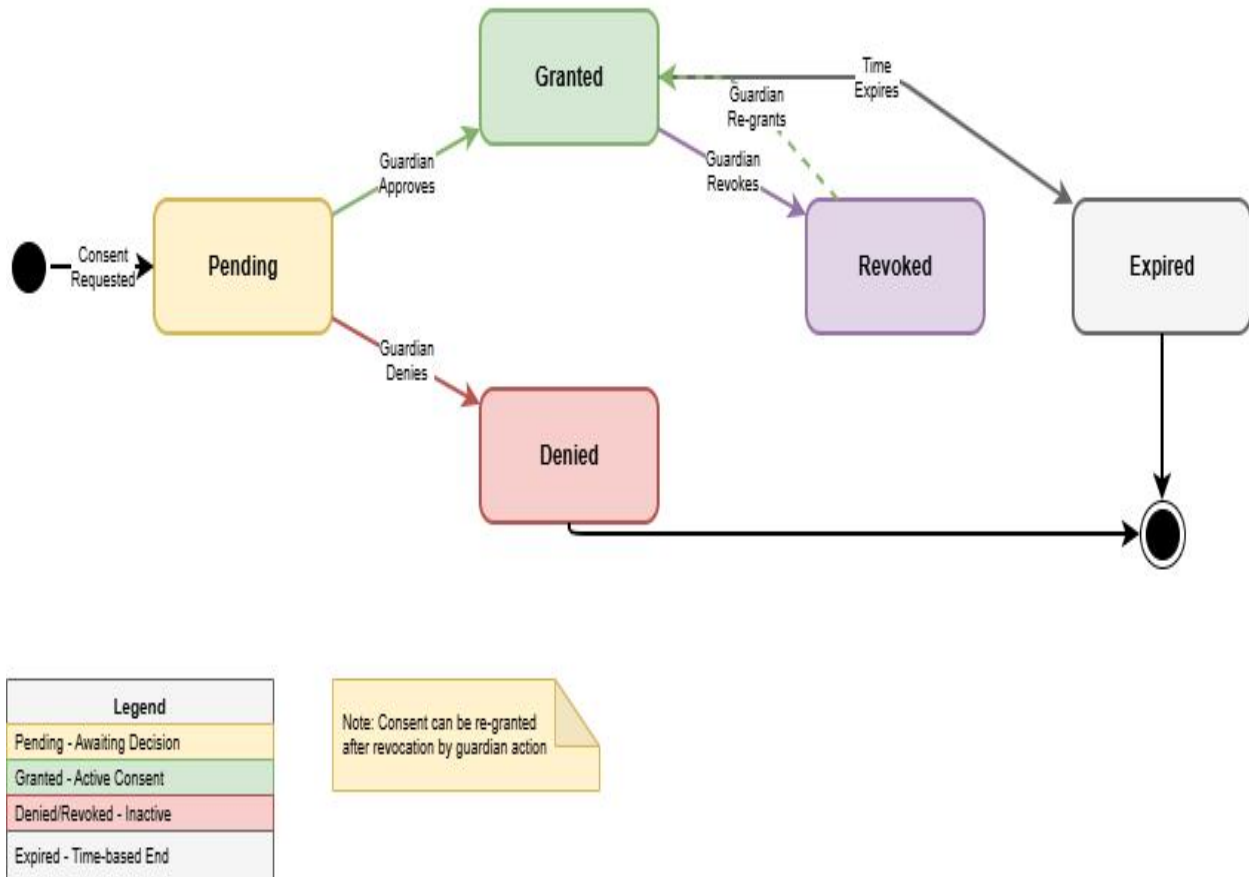


Figure 5 State Diagram – Consent Lifecycle

2.4.2.3 State Diagram – Alert Lifecycle

The alert lifecycle diagram represents the states an alert transitions through, from generation to acknowledgment and resolution. This ensures alerts remain informative and non-diagnostic.

Figure 2.11: State Diagram - Alert Lifecycle

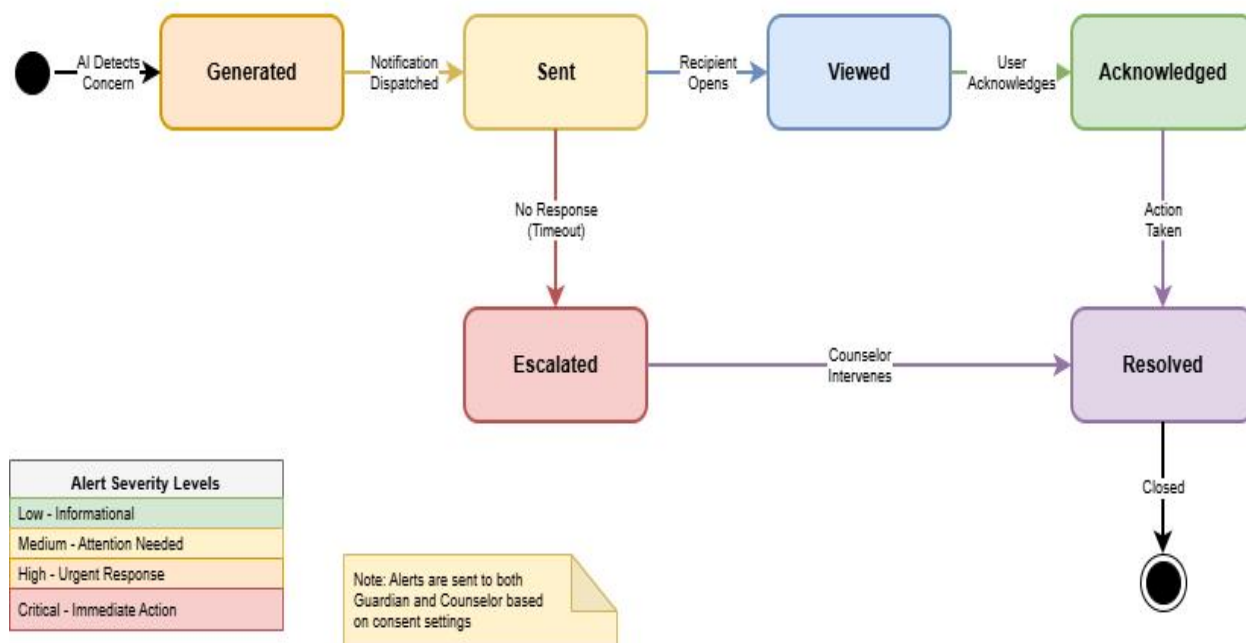


Figure 6 State Diagram – Alert Lifecycle

2.4.2 Process View

The Process View focuses on the **dynamic behavior** of the NEURONET system at runtime. It illustrates how system components collaborate over time to fulfill functional requirements, handle concurrency, and manage background processing such as AI analysis.

2.4.2.1 Sequence Diagram – Journal Submission and AI Analysis

This sequence diagram illustrates the interaction flow when an adolescent submits a journal entry, including authentication, data storage, asynchronous AI analysis, and insight generation.

Figure 2.13: Sequence Diagram - Journal Submission and AI Analysis

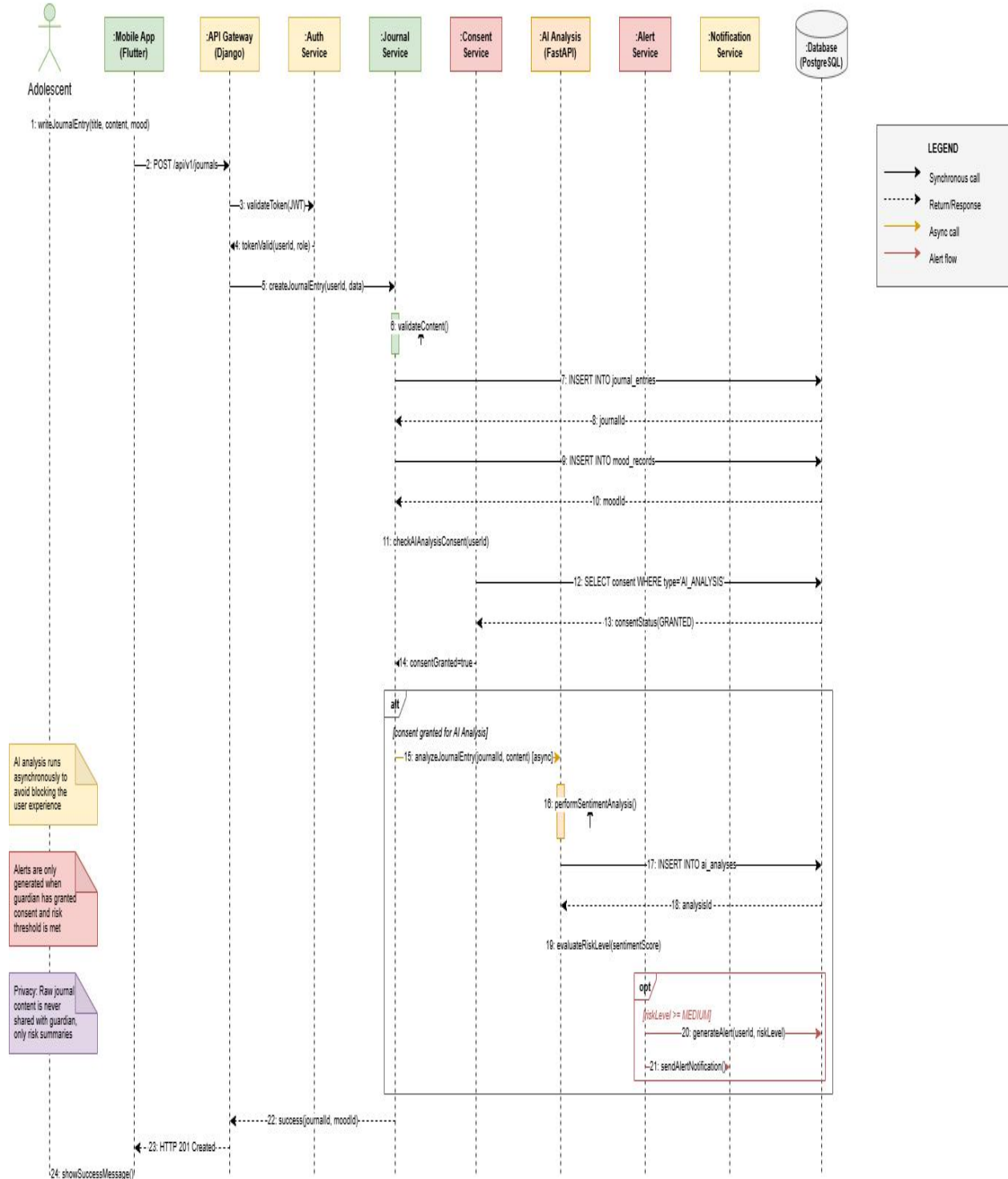


Figure 7 Sequence Diagram – Journal Submission and AI Analysis

2.4.2.2 Sequence Diagram – Alert Generation and Notification

This diagram shows how AI-generated indicators trigger alerts, how consent is validated, and how notifications are delivered to authorized users.

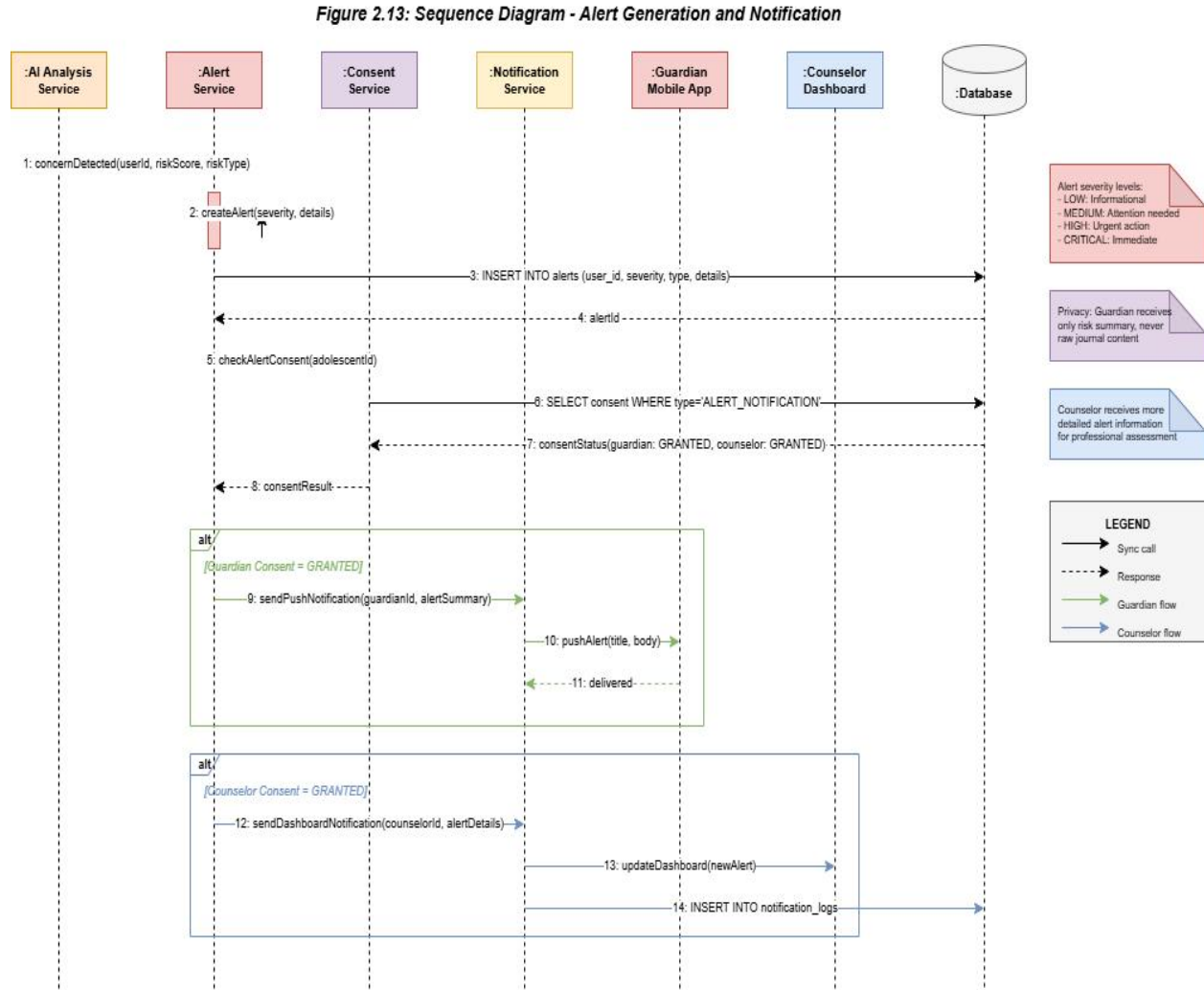


Figure 8 Sequence Diagram – Alert Generation and Notification

2.4.2.3 Sequence Diagram: Counselor-Adolescent Chat Initiation

Figure 2.14 illustrates the sequence of interactions required to initiate a chat session between a counselor and an adolescent user. This diagram emphasizes the privacy-first approach of the NEURONET system, where guardian consent is mandatory before any direct communication can occur.

Figure 2.14: Sequence Diagram - Counselor-Adolescent Chat Initiation

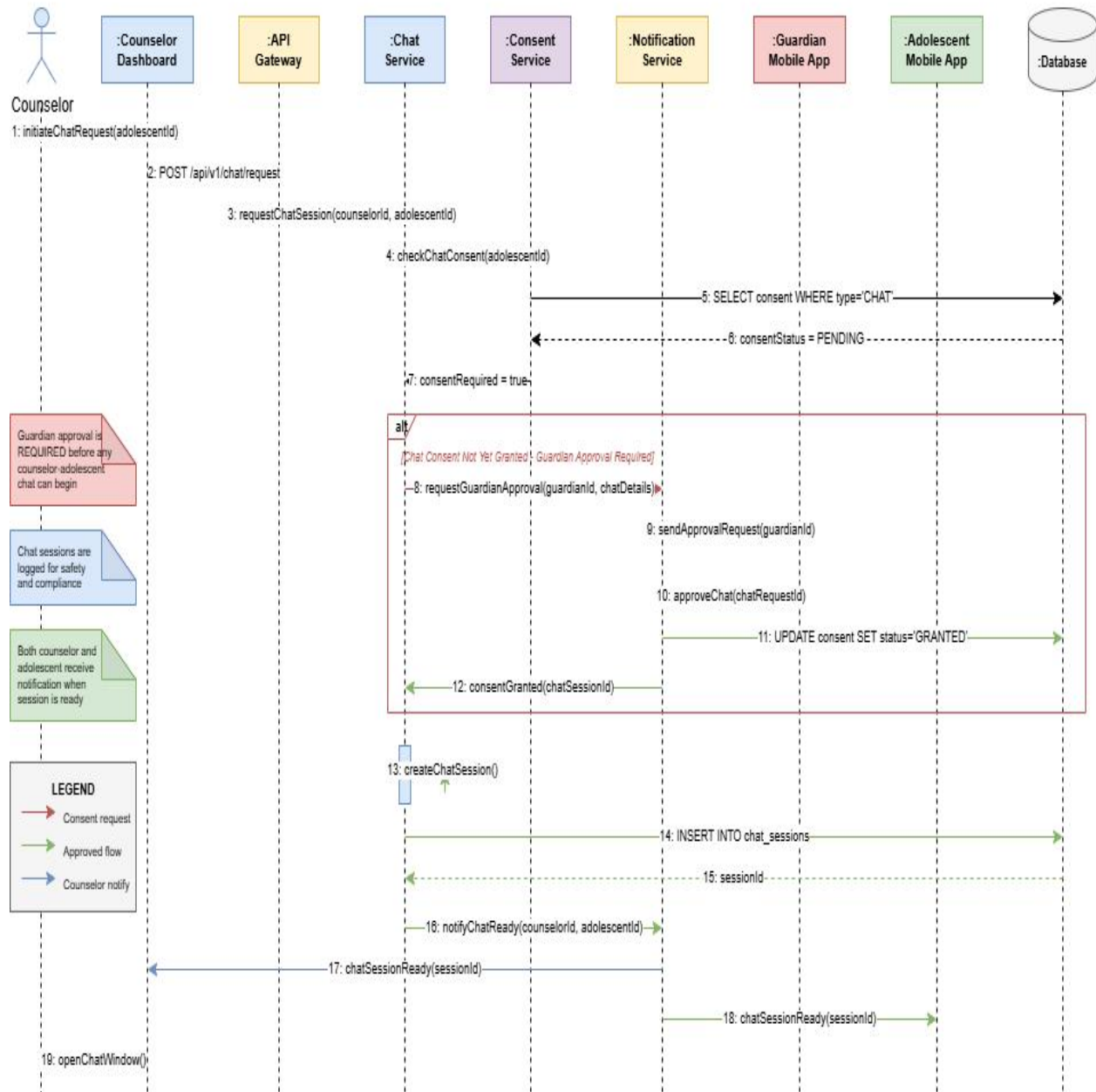


Figure 9 Sequence Diagram: Counselor-Adolescent Chat Initiation

2.4.2.4 Activity Diagram - User Registration Process

Figure 2.15: Activity Diagram - User Registration Process

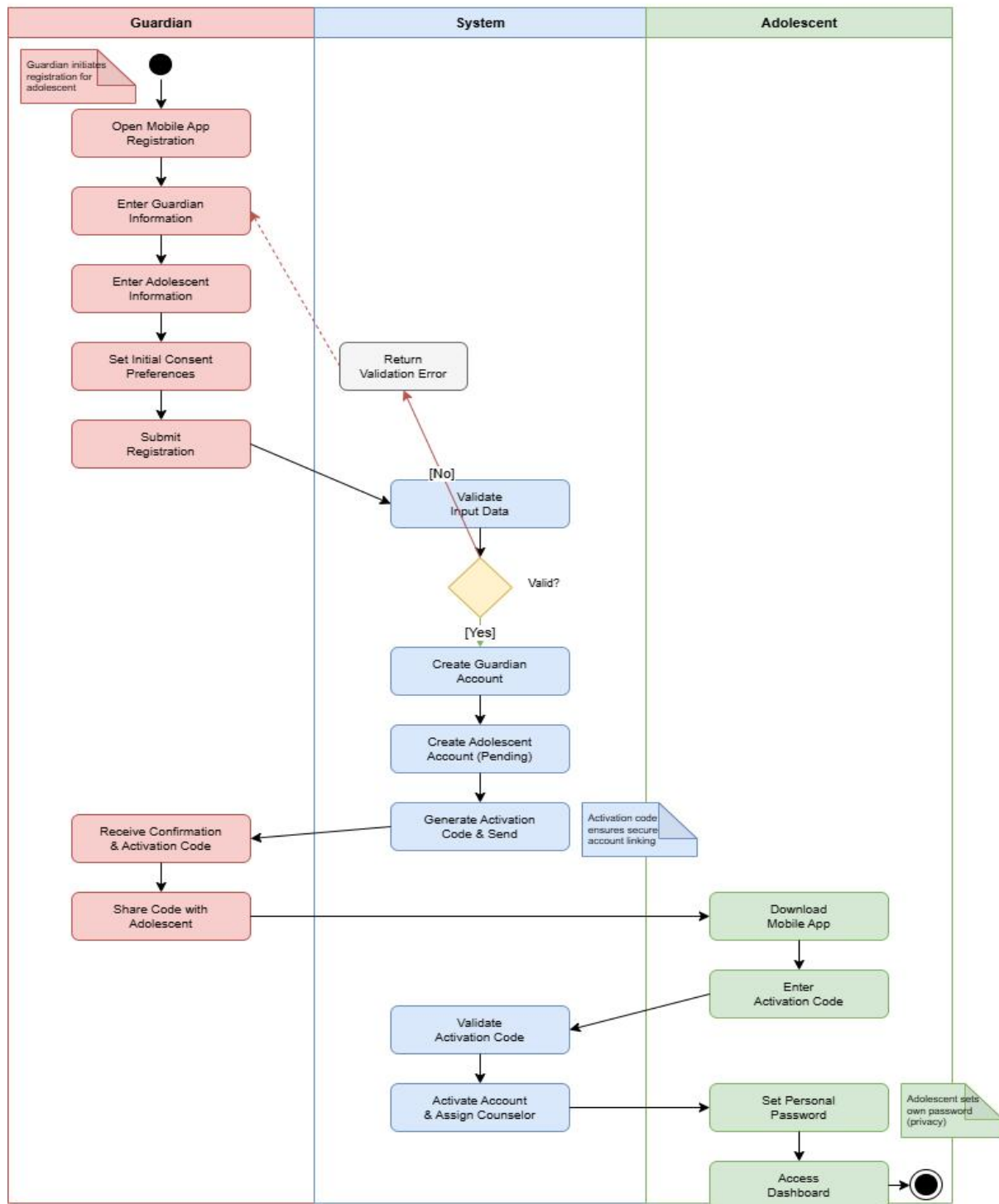


Figure 10 Activity Diagram - User Registration Process for guardian and adolescent

Figure 2.15b: Activity Diagram - Staff Registration (Web Portal)

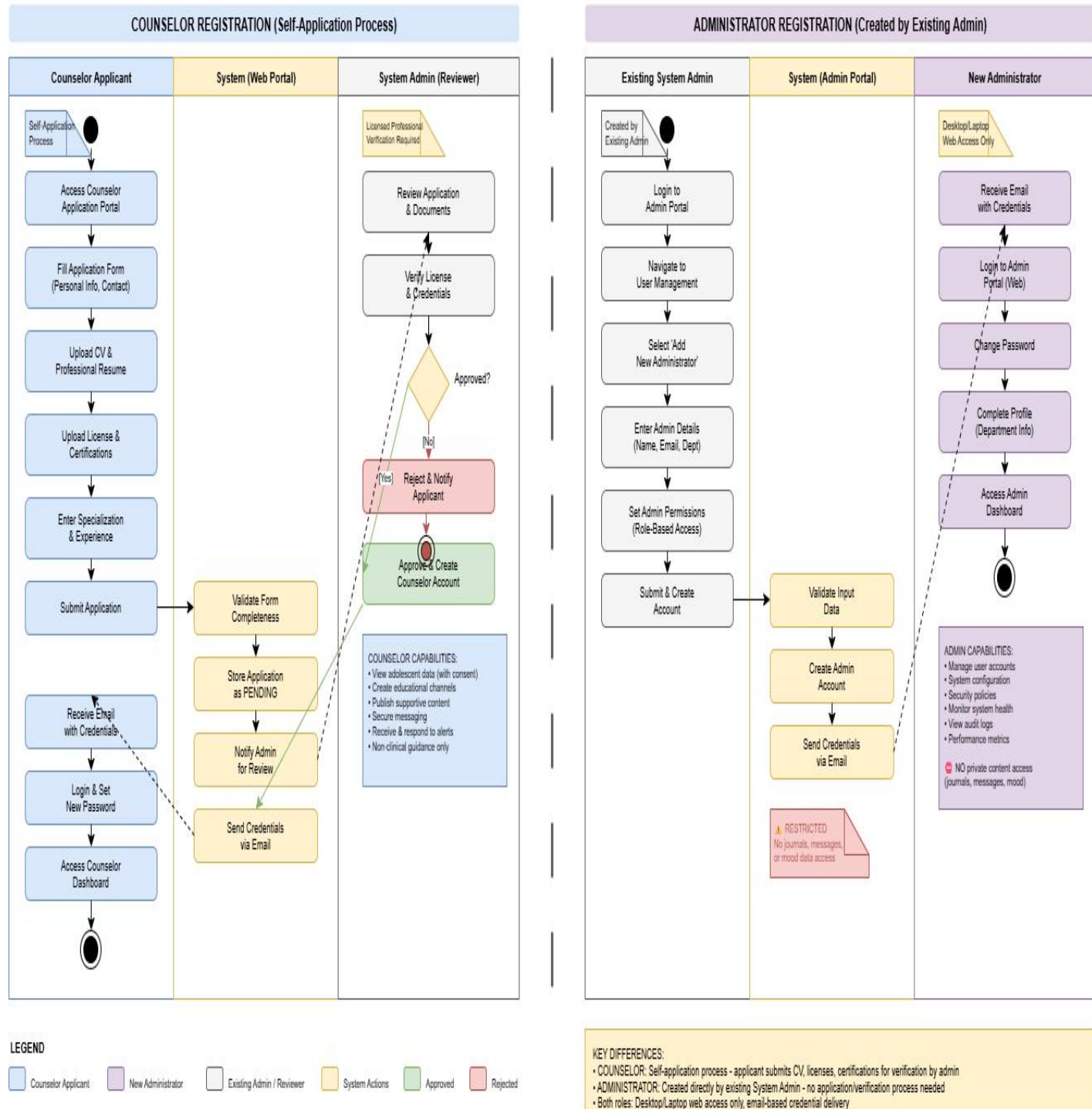


Figure 11 Activity Diagram - User Registration Process for counselor and admin through web portal

2.4.2.5 Activity Diagram - Consent Management Workflow

Figure 2.16: Activity Diagram - Consent Management Workflow

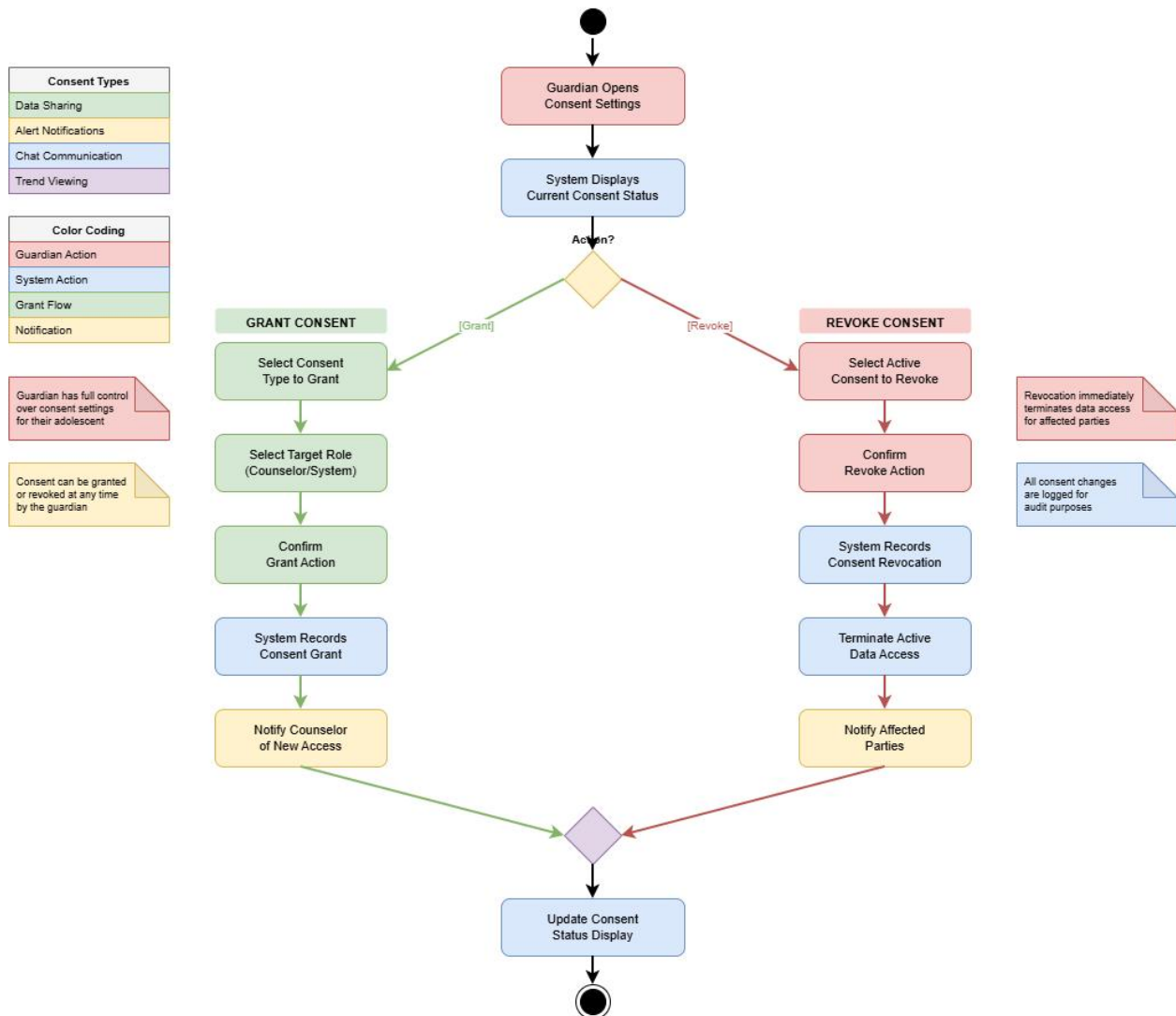


Figure 12 Activity Diagram - Consent Management Workflow

2.4.2.6 Activity Diagram – AI Analysis Process

The activity diagram represents the internal workflow of the AI analysis subsystem, including text preprocessing, sentiment scoring, trend evaluation, and output handling.

Figure 2.17: Activity Diagram - AI Analysis Process

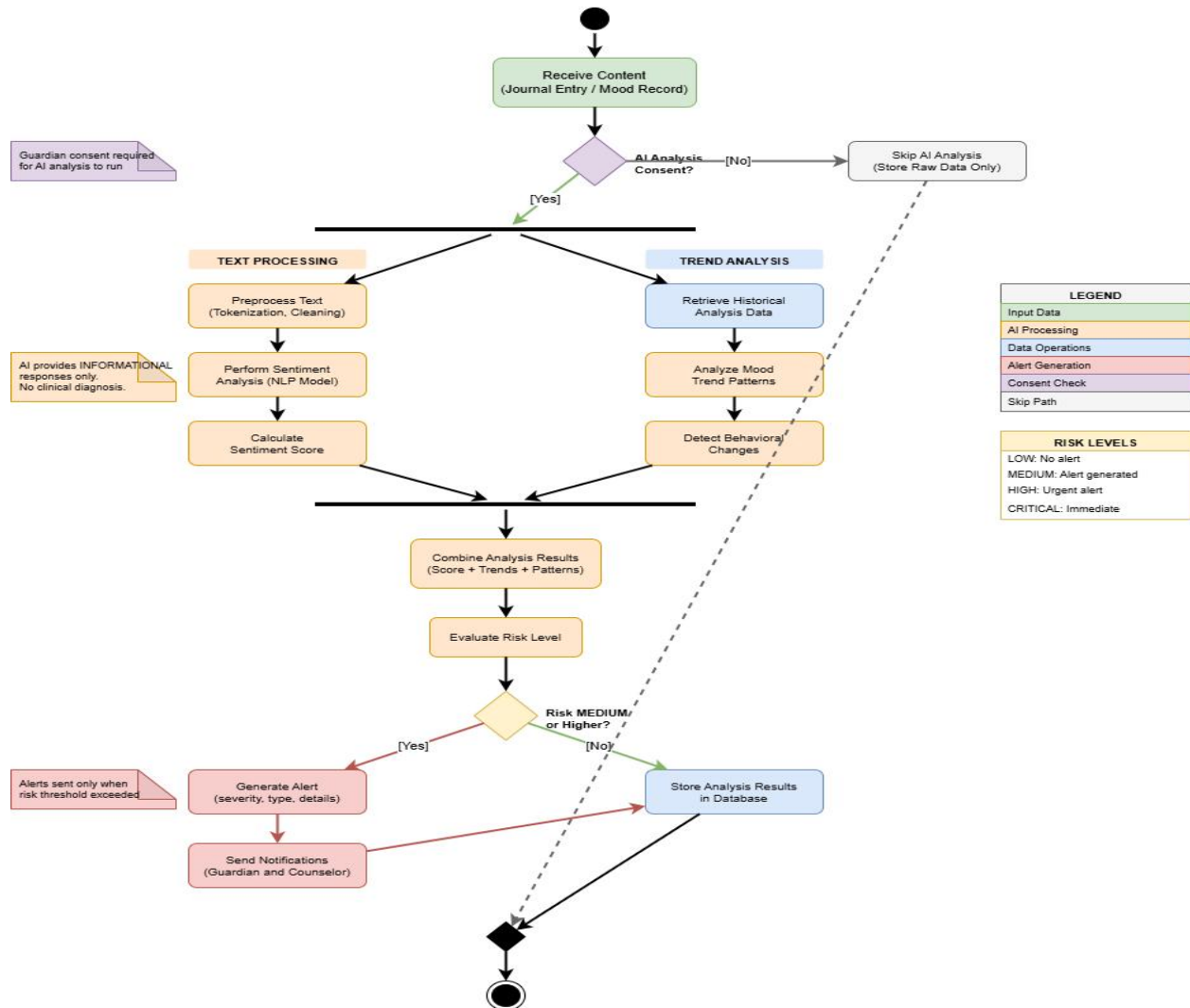


Figure 13 Activity Diagram – AI Analysis Process

2.4.3 Development View

The Development View presents the NEURONET system from an **implementation perspective**. It illustrates how the system is decomposed into modules, services, and components, supporting maintainability, modularity, and independent deployment.

2.4.3.1 Component Diagram – System Architecture

This diagram shows the organization of frontend applications, backend services, AI modules, and data storage components, along with their dependencies.

Figure 2.18: NEURONET System - Component Diagram

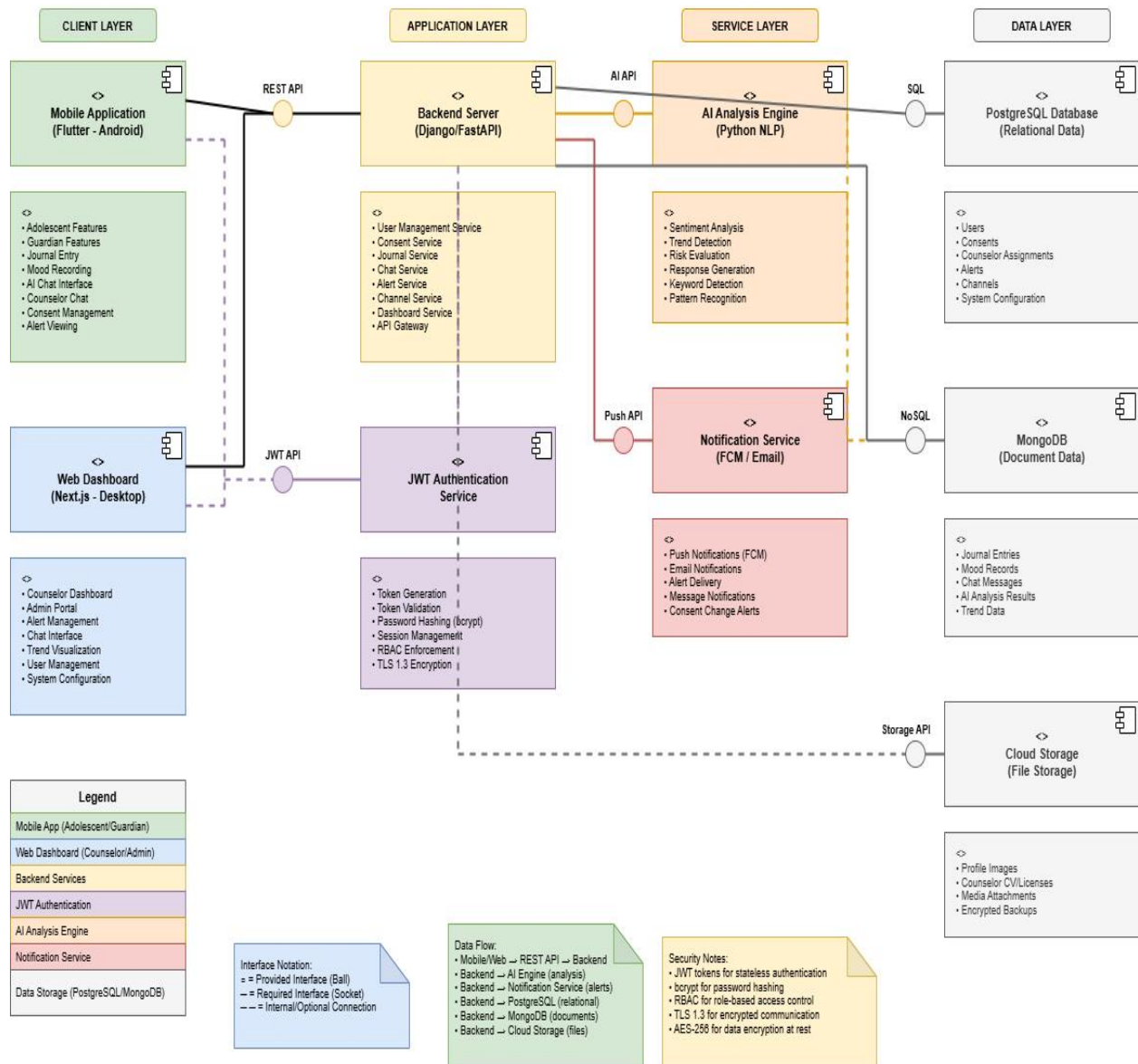


Figure 14 Component Diagram – System Architecture

2.4.4 Physical View

The Physical View describes how the NEURONET system is deployed on physical and cloud-based infrastructure. It focuses on runtime environments, hardware nodes, and communication pathways.

2.4.4.1 Deployment Diagram – Physical Infrastructure

The deployment diagram maps software components to cloud services, databases, and client devices, illustrating secure communication and infrastructure separation.

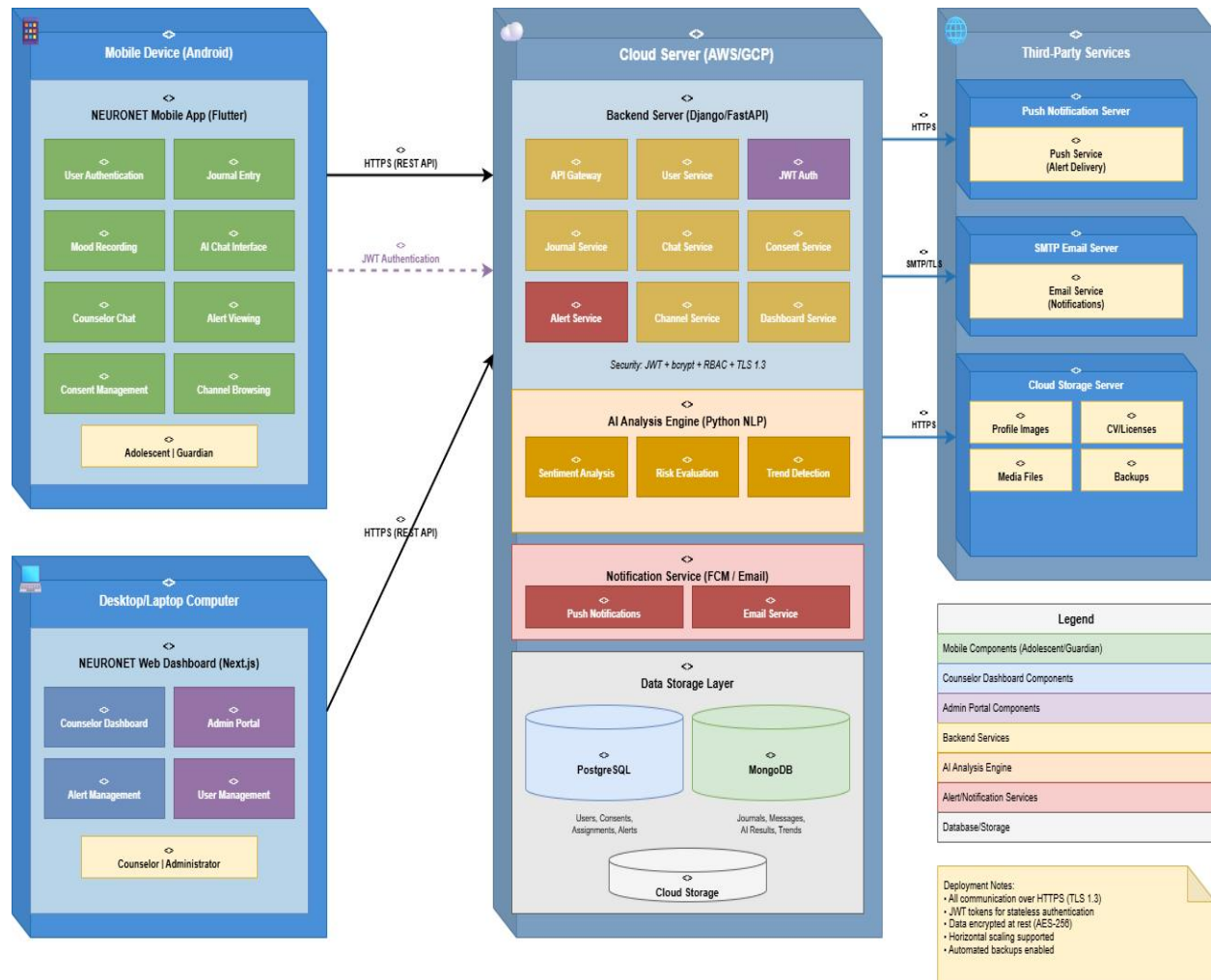


Figure 2.19: Deployment Diagram - NEURONET Physical Infrastructure

Figure 15 Deployment Diagram – Physical Infrastructure

2.4.5 Summary of Architectural Views

The **4+1 Architecture View Model** provides a comprehensive and balanced representation of the NEURONET system:

- **Use Case View (+1):** Ensures alignment between system functionality and user requirements
- **Logical View:** Defines the internal structure and relationships of core system entities
- **Process View:** Explains runtime behavior and component interaction
- **Development View:** Supports maintainability and scalability through modular design
- **Physical View:** Describes secure and efficient deployment architecture

CHAPTER THREE

DATA DESIGN

3.1 Logical Data Model

3.1.1 Overview

The logical data model represents the conceptual organization of data within the NEURONET system, independent of any specific database technology or physical storage considerations. It defines the core entities, their attributes, relationships, and constraints required to support system functionality while maintaining privacy, integrity, and role-based access control.

This model is derived from the Entity-Relationship Diagram (ERD) specified in the Software Requirements Specification (SRS). It supports all primary system operations, including user management, journaling, mood tracking, AI-driven analysis, consent enforcement, alert generation, and controlled communication between system actors.

The NEURONET system employs a polyglot persistence architecture, utilizing both PostgreSQL (relational) and MongoDB (document-oriented) databases to optimize for different data access patterns and requirements.

PostgreSQL (Relational Database):

Used for structured data requiring ACID compliance, referential integrity, and complex queries. This includes user accounts, consent records, counselor assignments, alerts and notifications, channel metadata and subscriptions, and audit logs.

MongoDB (Document Database):

Used for semi-structured and content-heavy data with flexible schemas. This includes journal entries, mood records with embedded trend data, chat messages, AI analysis results, and channel posts and interactions.

3.1.2 Logical Data Model Diagram

The logical data model for NEURONET shows both PostgreSQL tables (for relational data) and MongoDB collections (for document data). Color coding is used in diagrams to distinguish between the two database systems.

Figure 3.1: Logical Data Model for NEURONET System

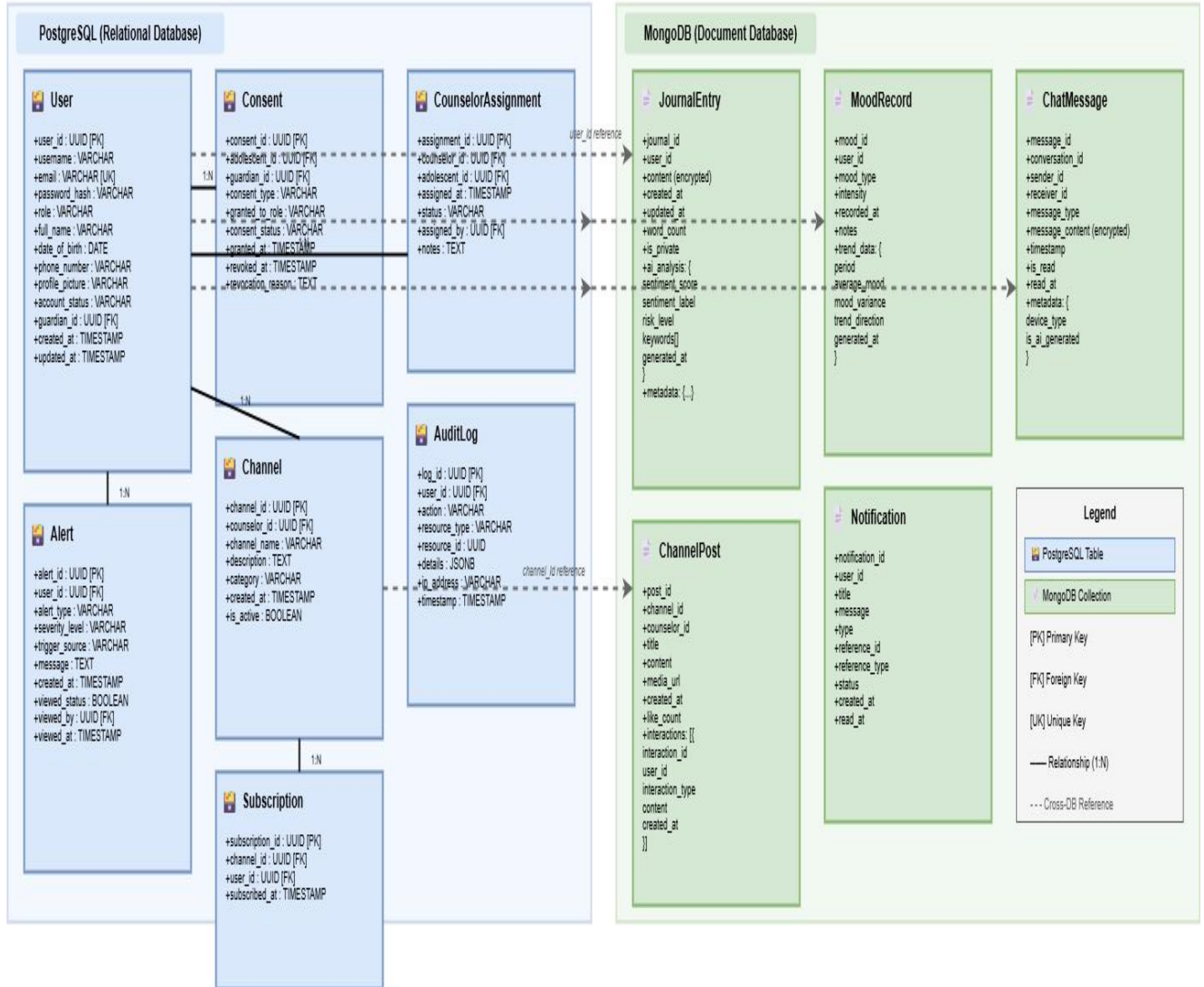


Figure 16 Logical Data Model for NEURONET System

3.1.3 PostgreSQL Entity Definitions (Relational Data)

User Table

Key Attributes:

- user_id (UUID, PK)
- full_name (VARCHAR(100), NOT NULL)
- email (VARCHAR(255), UNIQUE, NOT NULL)
- password_hash (VARCHAR(255), NOT NULL)
- role (VARCHAR(20), CHECK: adolescent, guardian, counselor, administrator)

- created_at (TIMESTAMP, DEFAULT CURRENT_TIMESTAMP)
- account_status (VARCHAR(20), DEFAULT 'active')

Relationships:

- One User can send many Chat Messages (as Sender)
- One User can receive many Chat Messages (as Receiver)

Guardian Table

Key Attributes:

- guardian_id (UUID, PK, FK → User.user_id)

Relationships:

- One Guardian → many GuardianAdolescentLinks
- One Guardian → many Consents

Adolescent Table

Key Attributes:

- adolescent_id (UUID, PK, FK → User.user_id)
- date_of_birth (DATE, NULLABLE)

Relationships:

- One Adolescent → many JournalEntries
- One Adolescent → many MoodRecords
- One Adolescent → many AIAalysis
- One Adolescent → many Alerts
- One Adolescent → many ChannelMemberships
- One Adolescent → many ChannelInteractions
- One Adolescent → many Consents
- One Adolescent → one active CounselorAssignment

Counselor Table

Key Attributes:

- counselor_id (UUID, PK, FK → User.user_id)
- professional_affiliation (VARCHAR, NULLABLE)
- institution_email (VARCHAR, NULLABLE)

- verification_status (VARCHAR(20), DEFAULT 'pending')

Relationships:

- One Counselor → many Channels
- One Counselor → many ChannelPosts
- One Counselor → many Alerts (acted upon)
- One Counselor → many CounselorAssignments

Admin Table

Key Attributes:

- admin_id (UUID, PK, FK → User.user_id)

Relationships:

- One Admin → many CounselorAssignments (assigned_by)

GuardianAdolescentLink Table

Key Attributes:

- link_id (UUID, PK)
- guardian_id (UUID, FK → Guardian.guardian_id)
- adolescent_id (UUID, FK → Adolescent.adolescent_id)
- relationship_type (VARCHAR)
- is_primary (BOOLEAN, DEFAULT FALSE)
- linked_at (TIMESTAMP)
- is_active (BOOLEAN)

Constraint:

- UNIQUE (adolescent_id) WHERE is_primary = TRUE

Relationships:

- One Guardian can have multiple Adolescent links
- One Adolescent can have multiple Guardian links

Consent Table

Key Attributes:

- consent_id (UUID, PK)
- adolescent_id (UUID, FK → Adolescent.adolescent_id)

- guardian_id (UUID, FK → Guardian.guardian_id)
- consent_type (VARCHAR)
- granted_to_role (VARCHAR)
- consent_status (VARCHAR)
- granted_at (TIMESTAMP)
- revoked_at (TIMESTAMP)
- expiry_date (TIMESTAMP, NULLABLE)
- notes (TEXT)

Relationships:

- Many Consent records belong to one User (Adolescent)
- Many Consent records are granted by one User (Guardian)

AIAnalysis Table

Key Attributes:

- analysis_id (UUID, PK)
- adolescent_id (UUID, FK → Adolescent.adolescent_id)
- analysis_type (VARCHAR)
- result_summary (TEXT)
- confidence_score (FLOAT)
- generated_at (TIMESTAMP)
- data_range_start (TIMESTAMP)
- data_range_end (TIMESTAMP)

Relationships:

- One AI Analysis Record belongs to one Adolescent

CounselorAssignment Table

Key Attributes:

- assignment_id (UUID, PK)
- counselor_id (UUID, FK → Counselor.counselor_id)
- adolescent_id (UUID, FK → Adolescent.adolescent_id)
- assigned_at (TIMESTAMP)
- assignment_status (VARCHAR)
- assigned_by (UUID, FK → Admin.admin_id)
- notes (TEXT)

Relationships:

- Many CounselorAssignment records link one Counselor to one Adolescent

- One Counselor can have many assignments
- One Adolescent has one active assignment at a time

Alert Table

Key Attributes:

- alert_id (UUID, PK)
- adolescent_id (UUID, FK → Adolescent.adolescent_id)
- alert_type (VARCHAR)
- severity_level (VARCHAR)
- trigger_description (TEXT)
- triggering_data (JSONB)
- created_at (TIMESTAMP)
- viewed_status (BOOLEAN)
- action_status (VARCHAR)
- action_taken_by (UUID, FK → Counselor.counselor_id)
- action_date (TIMESTAMP)
- action_notes (TEXT)

Relationships:

- Many Alert records belong to one User
- Alert can be viewed by Guardian or Counselor (with consent)

Channel Table

Key Attributes:

- channel_id (UUID, PK)
- counselor_id (UUID, FK → Counselor.counselor_id)
- channel_name (VARCHAR)
- description (TEXT)
- channel_type (VARCHAR)
- content_category (VARCHAR)
- created_at (TIMESTAMP)
- is_active (BOOLEAN)
- access_level (VARCHAR)
- allow_comments (BOOLEAN)
- allow_reactions (BOOLEAN)
- is_featured (BOOLEAN)

Relationships:

- One Counselor can create many Channels
- One Channel can have many ChannelPosts

- One Channel can have many ChannelMemberships

ChannelMembership Table

Key Attributes:

- membership_id (UUID, PK)
- channel_id (UUID, FK → Channel.channel_id)
- adolescent_id (UUID, FK → Adolescent.adolescent_id)
- joined_at (TIMESTAMP)
- membership_type (VARCHAR)
- notification_preference (VARCHAR)
- guardian_notified (BOOLEAN)

Relationships:

- Many Memberships belong to one Channel
- Many Memberships belong to one Adolescent

AuditLog Table

Key Attributes:

- **log_id:** UUID, Primary Key
- **user_id:** UUID, Foreign Key → User
- **action:** VARCHAR(100), Not Null
- **resource_type:** VARCHAR(50)
- **resource_id:** UUID
- **details:** JSONB
- **ip_address:** VARCHAR(45)
- **timestamp:** TIMESTAMP, Default CURRENT_TIMESTAMP

3.1.4 MongoDB Collection Definitions (Document Data)

JournalEntry Collection

Key Attributes:

- journal_id
- adolescent_id
- content (encrypted)
- created_at
- sentiment_score
- last_analyzed_at
- ai_metadata
- device_metadata

Relationships:

- Many JournalEntry documents belong to one User
- AI analysis is embedded within JournalEntry

MoodRecord Collection**Key Attributes:**

- mood_id
- adolescent_id
- mood_type
- intensity
- recorded_at
- context_notes
- trend_snapshot

Relationships:

- Many MoodRecord documents belong to one User
- Trend data is embedded within MoodRecord

ChatMessage Collection**Key Attributes:**

- message_id
- conversation_id
- sender_id
- receiver_id
- message_type (CounselorChat / GuardianChat / AdminAlert)
- message_content (encrypted)
- timestamp
- read_status
- metadata

Relationships:

- Many ChatMessage documents are sent by one User
- Many ChatMessage documents are received by one User
- Messages are grouped by conversation ID

ChannelPost Collection

Key Attributes:

- post_id
- channel_id
- counselor_id
- title
- content
- media_url
- created_at
- updated_at
- interactions (embedded summaries)

Relationships:

- Many ChannelPost documents belong to one Channel
- Interactions are embedded within ChannelPost

3.2 Physical Data Management

3.2.1 Database Architecture Overview

The NEURONET system adopts a **polyglot persistence architecture**, utilizing multiple database technologies optimized for different data types and operational requirements. This approach improves scalability, performance, and flexibility while preserving data integrity for critical system components.

The system employs the following database technologies:

- **PostgreSQL** – A relational database used for structured data requiring strong consistency, transactional integrity (ACID compliance), and relational constraints.
- **MongoDB** – A document-oriented database used for semi-structured and unstructured content with variable schemas and high write volumes.

3.2.2 Physical Database Schema

The physical database schema defines how logical entities are mapped to concrete database structures across PostgreSQL and MongoDB. It specifies tables, collections, attributes, data types, constraints, and relationships.

3.2.3 Data Storage Strategy

PostgreSQL Usage

PostgreSQL is used to store structured and relational data that requires strict integrity and controlled access:

- User accounts and authentication credentials
- Consent records with lifecycle enforcement
- Counselor-to-adolescent assignments
- Alert generation and status tracking
- Channel metadata and subscription relationships
- Mood records with enumerated values
- Audit logs for monitoring and compliance

MongoDB Usage

MongoDB is used to store content-heavy and flexible data structures:

- Journal entries containing variable-length encrypted text
- Chat messages with high-frequency, time-series characteristics
- AI analysis results with extensible schemas
- Channel interactions including reactions and comments

3.2.4 Data Security Measures

Encryption at Rest

- PostgreSQL employs Transparent Data Encryption (TDE)
- MongoDB uses an encrypted storage engine

Encryption in Transit

- TLS 1.3 is enforced for all database connections
- Secure, encrypted connection strings are used across services

Access Control

- Database-level role-based access control
- Application-level least-privilege credentials
- Segregation of read and write permissions where applicable

Backup and Recovery

- Automated daily backups
- Point-in-time recovery capabilities

- Encrypted backup storage
- Regular backup integrity verification

3.3 Data Dictionary

3.3.1 Overview

The data dictionary provides a detailed description of all data elements used within the NEURONET system. It serves as a reference for developers, database administrators, and evaluators by defining the structure, meaning, constraints, and usage of each attribute stored in the system.

This section documents both relational tables stored in PostgreSQL and document-based collections stored in MongoDB. The data dictionary ensures consistency between the logical data model and the physical database implementation while supporting data integrity, security, and scalability requirements.

3.3.2 PostgreSQL Tables

Table 3 User Table Structure

Attribute	Data Type	Size	Constraints	Description
user_id	UUID	36	PK, NOT NULL	Unique user identifier
full_name	VARCHAR	100	NOT NULL	User's full name
email	VARCHAR	255	UNIQUE, NOT NULL	Email address
password_hash	VARCHAR	255	NOT NULL	Bcrypt-hashed password
role	VARCHAR	20	NOT NULL	Adolescent / Guardian / Counselor / Administrator
created_at	TIMESTAMP	–	DEFAULT NOW()	Account creation time
account_status	VARCHAR	20	DEFAULT 'active'	Active / Inactive / Suspended

Table 4 Guardian Table Structure

Attribute	Data Type	Size	Constraints	Description
guardian_id	UUID	36	PK, FK → User.user_id	Guardian user reference

Table 5 Adolescent Table Structure

Attribute	Data Type	Size	Constraints	Description
adolescent_id	UUID	36	PK, FK → User.user_id	Adolescent user reference
date_of_birth	DATE	–	NULLABLE	Birth date

Table 6 Counselor Table Structure

Attribute	Data Type	Size	Constraints	Description
counselor_id	UUID	36	PK, FK → User.user_id	Counselor user reference
professional_affiliation	VARCHAR	100	NULLABLE	Organization or affiliation
institution_email	VARCHAR	255	NULLABLE	Institutional email
verification_status	VARCHAR	20	DEFAULT 'pending'	Pending / Approved / Rejected

Table 7 GuardianAdolescentLink Table Structure

Attribute	Data Type	Size	Constraints	Description
link_id	UUID	36	PK	Link identifier
guardian_id	UUID	36	FK → Guardian.guardian_id	Guardian reference
adolescent_id	UUID	36	FK → Adolescent.adolescent_id	Adolescent reference
relationship_type	VARCHAR	30	–	Parent / LegalGuardian / Other
is_primary	BOOLEAN	–	DEFAULT FALSE	Primary guardian flag
linked_at	TIMESTAMP	–	DEFAULT NOW()	Link creation time
is_active	BOOLEAN	–	DEFAULT TRUE	Link status

Table 8 Consent Table Structure

Attribute	Data Type	Size	Constraints	Description
consent_id	UUID	36	PK	Consent identifier
adolescent_id	UUID	36	FK, NOT NULL	Adolescent reference
guardian_id	UUID	36	FK, NOT NULL	Granting guardian (primary only)
consent_type	VARCHAR	50	NOT NULL	JournalAnalysis / DataSharing / Communication
granted_to_role	VARCHAR	20	NOT NULL	Guardian / Counselor / Both
consent_status	VARCHAR	20	NOT NULL	Granted / Revoked / Expired
granted_at	TIMESTAMP	–	NOT NULL	Grant timestamp
revoked_at	TIMESTAMP	–	NULLABLE	Revocation timestamp
expiry_date	TIMESTAMP	–	NULLABLE	Expiry date
notes	TEXT	–	NULLABLE	Revocation reason

Table 9 CounselorAssignment Table Structure

Attribute	Data Type	Size	Constraints	Description
assignment_id	UUID	36	PK	Assignment identifier
counselor_id	UUID	36	FK, NOT NULL	Counselor reference
adolescent_id	UUID	36	FK, NOT NULL	Adolescent reference
assigned_at	TIMESTAMP	–	DEFAULT NOW()	Assignment time
assignment_status	VARCHAR	20	DEFAULT 'active'	Active / Inactive / Pending

Attribute	Data Type	Size	Constraints	Description
assigned_by	UUID	36	FK → Admin.admin_id	Assigning admin
notes	TEXT	–	NULLABLE	Assignment reason

Table 10 AIAAnalysis Table Structure

Attribute	Data Type	Size	Constraints	Description
analysis_id	UUID	36	PK	Analysis identifier
adolescent_id	UUID	36	FK, NOT NULL	Adolescent reference
analysis_type	VARCHAR	30	NOT NULL	Sentiment / Trend / Pattern
result_summary	TEXT	–	–	Human-readable outcome
confidence_score	FLOAT	–	–	Model confidence (0–1)
data_range_start	TIMESTAMP	–	–	Analysis start period
data_range_end	TIMESTAMP	–	–	Analysis end period
generated_at	TIMESTAMP	–	DEFAULT NOW()	Generation timestamp

Table 11 Alert Table Structure

Attribute	Data Type	Size	Constraints	Description
alert_id	UUID	36	PK	Alert identifier
adolescent_id	UUID	36	FK, NOT NULL	Adolescent reference
alert_type	VARCHAR	50	NOT NULL	MoodDrop / Pattern / ContentFlag
severity_level	VARCHAR	20	NOT NULL	Low / Medium / High
trigger_description	TEXT	–	–	Trigger explanation
triggering_data	JSONB	–	NULLABLE	Trigger data snapshot
created_at	TIMESTAMP	–	DEFAULT NOW()	Creation time
viewed_status	BOOLEAN	–	DEFAULT FALSE	View status
action_status	VARCHAR	20	DEFAULT 'Pending'	Pending / Reviewed / Resolved
action_taken_by	UUID	36	FK → Counselor.counselor_id	Acting counselor
action_date	TIMESTAMP	–	NULLABLE	Action timestamp

Table 12 Channel Table Structure

Attribute	Data Type	Size	Constraints	Description
channel_id	UUID	36	PK	Channel identifier
counselor_id	UUID	36	FK, NOT NULL	Creator counselor
channel_name	VARCHAR	100	NOT NULL	Channel name
description	TEXT	–	–	Channel description
channel_type	VARCHAR	30	–	Educational / Support
created_at	TIMESTAMP	–	DEFAULT NOW()	Creation time
is_active	BOOLEAN	–	DEFAULT TRUE	Status

Table 13 MoodRecord Table Structure

Attribute	Data Type	Size	Constraints	Description
mood_id	UUID	36	PK, NOT NULL	Unique identifier
user_id	UUID	36	FK, NOT NULL	User reference
mood_type	VARCHAR	30	NOT NULL	Mood category (Happy, Sad, Anxious, Calm, Stressed, Neutral, etc.)
recorded_at	TIMESTAMP	-	DEFAULT NOW()	Recording timestamp

Table 14 AuditLog Table Structure

Attribute	Data Type	Size	Constraints	Description
log_id	UUID	36	PK, NOT NULL	Unique identifier
user_id	UUID	36	FK	Actor reference
action	VARCHAR	100	NOT NULL	Action performed
resource_type	VARCHAR	50	-	Resource category
resource_id	UUID	36	-	Resource reference
timestamp	TIMESTAMP	-	DEFAULT NOW()	Action timestamp
ip_address	VARCHAR	45	-	Client IP address

3.3.3 MongoDB Collections

Table 15 JournalEntry Collection Structure

Field	Data Type	Constraints	Description
_id	ObjectId	PK	MongoDB document ID
journal_id	UUID	NOT NULL	Unique journal identifier
user_id	UUID	NOT NULL, INDEX	Adolescent user reference
content	String	NOT NULL	Encrypted journal text used as AI input
created_at	ISODate	NOT NULL, INDEX	Journal creation timestamp
sentiment_score	Number	Range 0–1	AI-generated overall sentiment score
sentiment_confidence	Number	Range 0–1	Confidence level of AI sentiment result
word_count	Number	≥ 0	Entry word count used for AI constraints

Field	Data Type	Constraints	Description
language_detected	String	DEFAULT en	Language detected by NLP model
analysis_status	String	DEFAULT pending	Status (Pending, Completed, Failed)
retention_expires_at	ISODate	NOT NULL	Automatic deletion date per policy
deleted_at	ISODate	NULL	Soft-deletion timestamp
metadata	Object	—	Additional metadata
↳ device_type	String	—	Device used
↳ app_version	String	—	Application version

Table 16 ChatMessage Collection Structure

Field	Data Type	Constraints	Description
id	ObjectId	PK	MongoDB document ID
message_id	UUID	NOT NULL	Unique message identifier
sender_id	UUID	NOT NULL, INDEX	Sender reference
receiver_id	UUID	NOT NULL, INDEX	Receiver reference
message_content	String	NOT NULL	Encrypted message text
timestamp	ISODate	NOT NULL, INDEX	Message timestamp
message_type	String	NOT NULL	Type (AI_Chat, Counselor_Chat, Guardian_Chat)
is_read	Boolean	DEFAULT FALSE	Read status
metadata	Object	-	Additional metadata
↳ conversation_id	UUID	-	Conversation grouping

Table 17 AIAnalysis Collection Structure

Field	Data Type	Constraints	Description
id	ObjectId	PK	MongoDB document ID
analysis_id	UUID	NOT NULL	Unique analysis identifier
user_id	UUID	NOT NULL, INDEX	User reference
analysis_type	String	NOT NULL	Type (Sentiment, Trend)
source_type	String	NOT NULL	Source (Journal, Chat)
source_id	UUID	NOT NULL	Source document reference
result_summary	String	-	Human-readable summary
sentiment_scores	Object	-	Detailed scores
↳ positive	Number	-	Positive sentiment (0-1)
↳ negative	Number	-	Negative sentiment (0-1)
↳ neutral	Number	-	Neutral sentiment (0-1)
keywords	Array[String]	-	Detected keywords
generated_at	ISODate	NOT NULL, INDEX	Analysis timestamp

Table 18 ChannelInteraction Collection Structure

Field	Data Type	Constraints	Description
id	ObjectId	PK	MongoDB document ID
interaction_id	UUID	NOT NULL	Unique interaction ID
channel_id	UUID	NOT NULL, INDEX	Channel reference
user_id	UUID	NOT NULL, INDEX	User reference
interaction_type	String	NOT NULL	Type (Reaction, Comment)
content	String	-	Interaction content
created_at	ISODate	NOT NULL	Interaction timestamp

3.3.4 Enumeration Values

User Roles:

- Adolescent
- Guardian
- Counselor
- Administrator

Account Status:

- Active
- Inactive
- Suspended
- Pending_Activation

Consent Status:

- Granted
- Revoked
- Expired

Alert Severity:

- Low
- Medium
- High

Mood Types:

- Happy
- Sad
- Anxious
- Calm
- Stressed
- Neutral
- Excited
- Tired
- Angry
- Hopeful

Message Types:

- AI_Chat
- Counselor_Chat
- Guardian_Chat

Interaction Types:

- Reaction
- Comment

3.3.5 Index Definitions**PostgreSQL Indexes:**

- users_email_idx ON users(email)
- users_role_idx ON users(role)
- consents_adolescent_idx ON consents(adolescent_id)
- alerts_user_idx ON alerts(user_id)
- alerts_created_idx ON alerts(created_at)
- assignments_counselor_idx ON counselor_assignments(counselor_id)

MongoDB Indexes:

- journal_entries: { user_id: 1, created_at: -1 }
- chat_messages: { sender_id: 1, timestamp: -1 }
- chat_messages: { receiver_id: 1, timestamp: -1 }
- ai_analyses: { user_id: 1, generated_at: -1 }
- channel_interactions: { channel_id: 1, created_at: -1 }

CHAPTER FOUR

HUMAN INTERFACE DESIGN

4.1 Overview

4.1.1 Purpose of Human Interface Design

This chapter describes the user interface design for the NEURONET system, providing visual representations and functional descriptions of the screens that users will interact with. The interface design ensures that the system is accessible, intuitive, and appropriate for all user roles while maintaining the ethical and privacy constraints defined in the SRS.

4.1.2 Design Principles

The NEURONET user interface is designed according to the following principles:

User-Centered Design

Interfaces are tailored to each user role's needs and technical proficiency. Adolescent interfaces prioritize simplicity and emotional safety, while professional interfaces (counselors and administrators) provide comprehensive functionality.

Consistency

Uniform navigation patterns are used across platforms with consistent visual language, terminology, and predictable interaction behaviors.

Accessibility

The interface uses readable fonts, appropriate color contrasts, touch-friendly elements for mobile use, and a clear visual hierarchy.

Privacy Awareness

Data visibility and sharing are clearly indicated. Consent status is prominently displayed, and private content is protected from unauthorized access.

Supportive Communication

Non-judgmental language, encouraging messaging, and non-alarming presentation of alerts and insights are used throughout the system.

4.1.3 Platform Overview

The NEURONET system supports multiple platforms to accommodate different user roles:

Mobile Application (Flutter)

Designed primarily for adolescent users, optimized for Android devices, and focused on journaling, mood tracking, and AI interaction.

Web Dashboard (Next.js)

Designed for guardians, counselors, and administrators with responsive layouts and role-specific views.

4.1.4 User Role Interface Summary

Table 19 Interface Summary by User Role

User Role	Platform	Primary Functions
Adolescent	Mobile Application	Journal entry, mood recording, AI chat, personal dashboard, counselor chat, channel browsing
Guardian	Mobile Application	Adolescent registration, consent management, aggregated insights, alerts, counselor communication
Counselor	Web Dashboard	Assigned adolescents view, trend analysis, alert management, communication, channel management
Administrator	Web Dashboard	User management, counselor assignments, system monitoring, configuration

4.2 Screen Images

This section presents the user interface designs for the NEURONET system, showcasing both mobile and web-based interfaces. The mobile application screens are designed for adolescents and guardians, providing intuitive access to journaling, AI support, and monitoring features. The web-based dashboards are tailored for counselors and system administrators, offering comprehensive tools for patient management, alert monitoring, and system oversight. Each screen has been carefully designed with user experience in mind, incorporating modern UI principles, accessibility standards, and privacy-first design patterns. The color schemes differentiate user roles while maintaining visual consistency across the platform: green for adolescent features, pink for guardian interfaces, blue for counselor dashboards, and purple for administrative functions.

The following screens demonstrate the complete user journey across different stakeholder roles within the NEURONET ecosystem.

MOBILE APPLICATION SCREENS

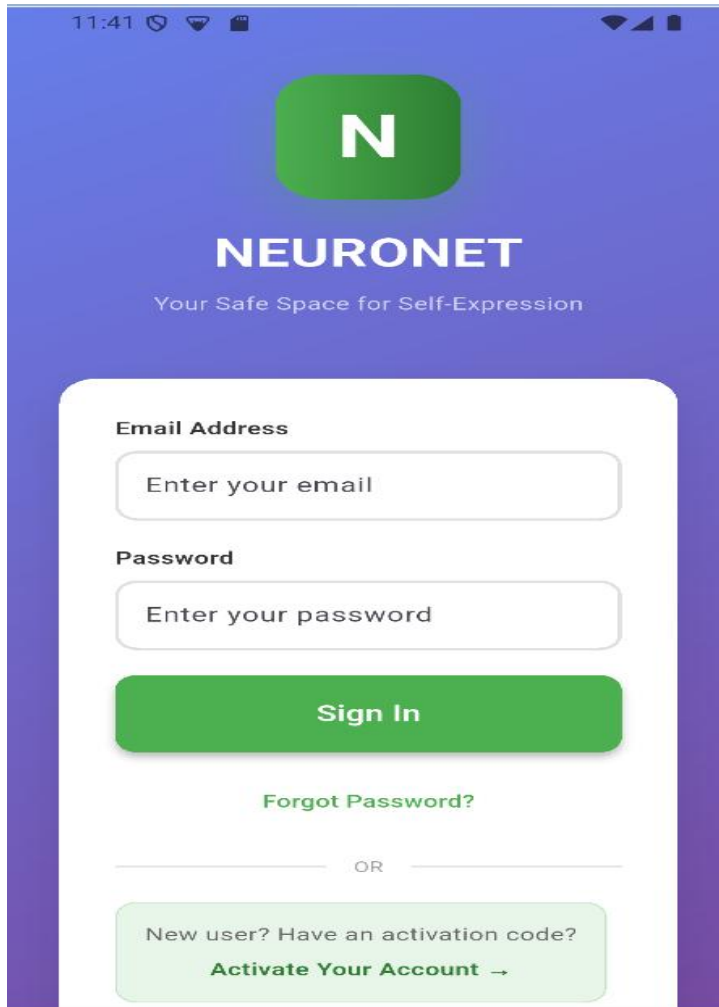


Figure 17 Adolescent Login Screen

Description: The login interface for adolescent users, featuring email and password authentication, password recovery option, and account activation for new users with activation codes.

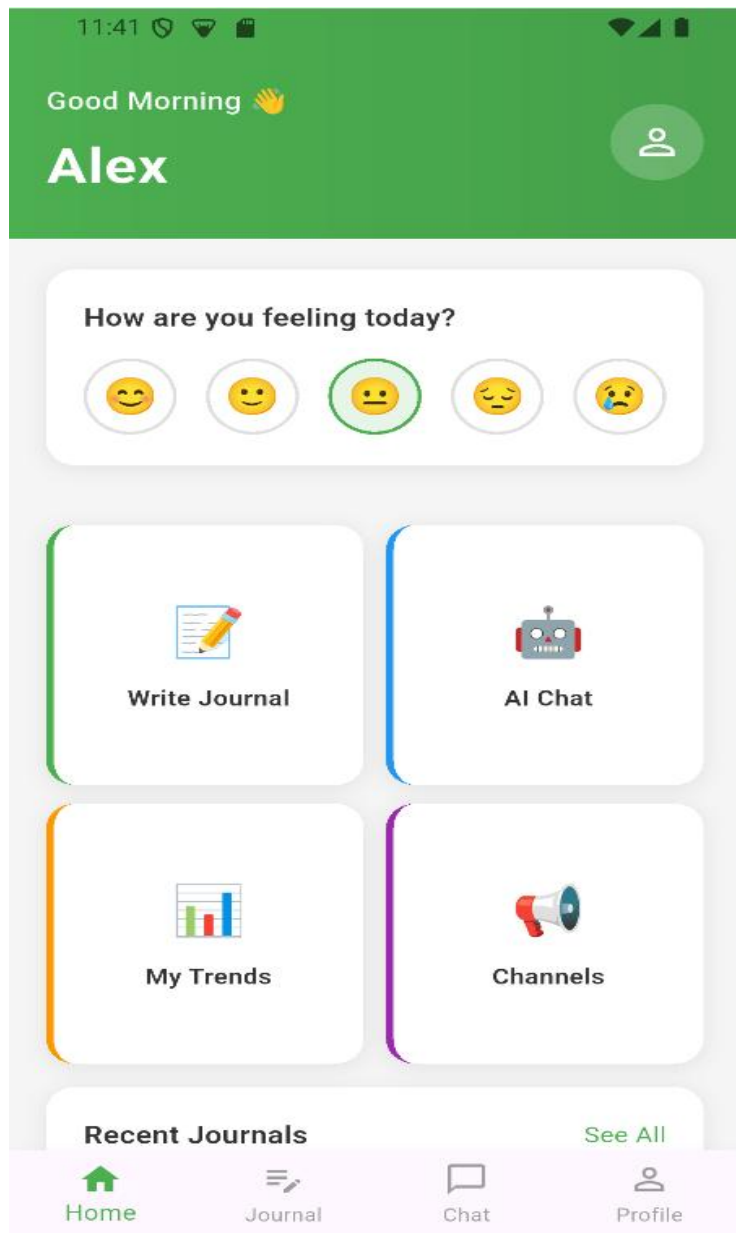


Figure 18 Adolescent Dashboard Screen

Figure 4.2: Adolescent Dashboard Screen

Description: The main dashboard for adolescent users, displaying a personalized greeting, mood selector with five emotional states, quick access cards for journaling and AI chat, and a summary of recent journal entries.

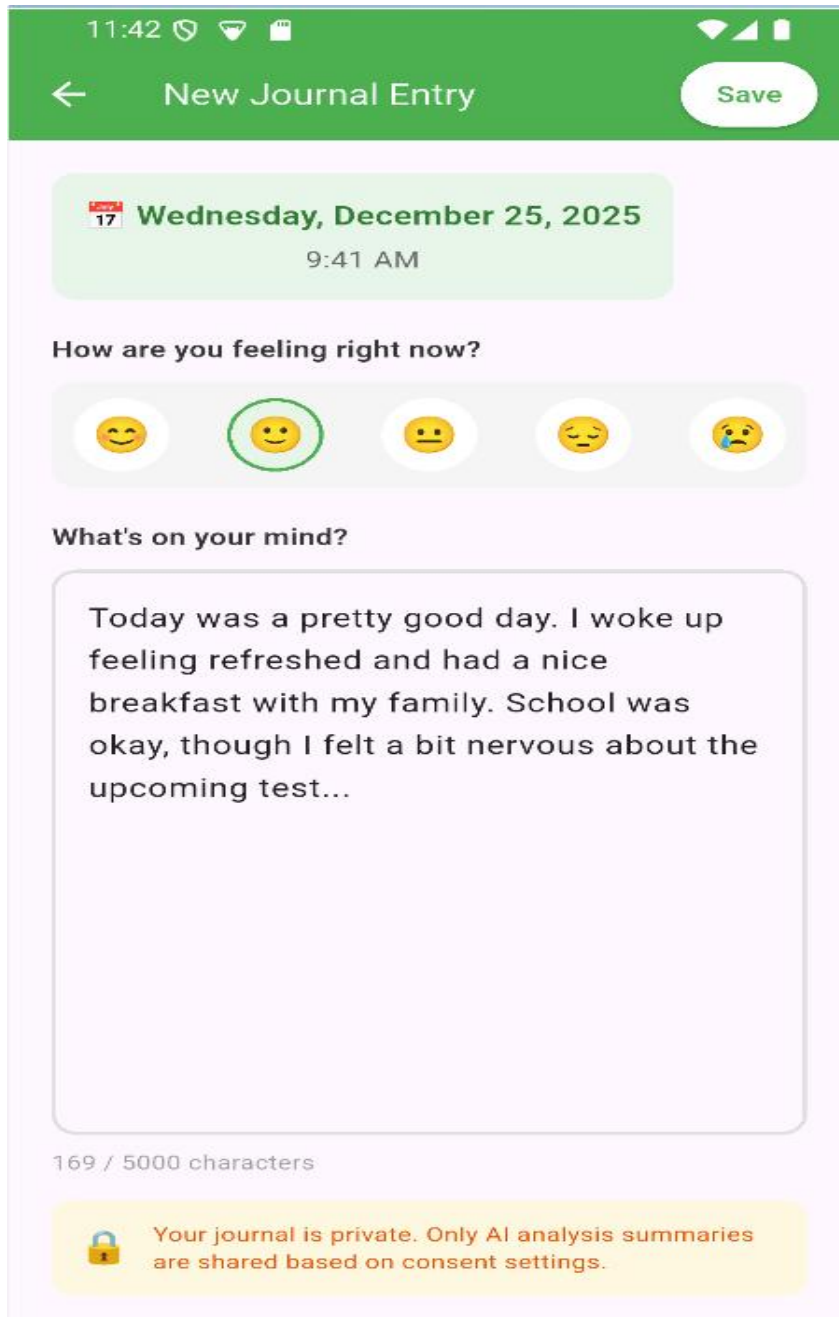


Figure 19 Journal Entry Screen

Description: The journaling interface where adolescents can record their thoughts and feelings, featuring date/time display, mood selection, a text area with character counter (5000 character limit), and privacy assurance notice.

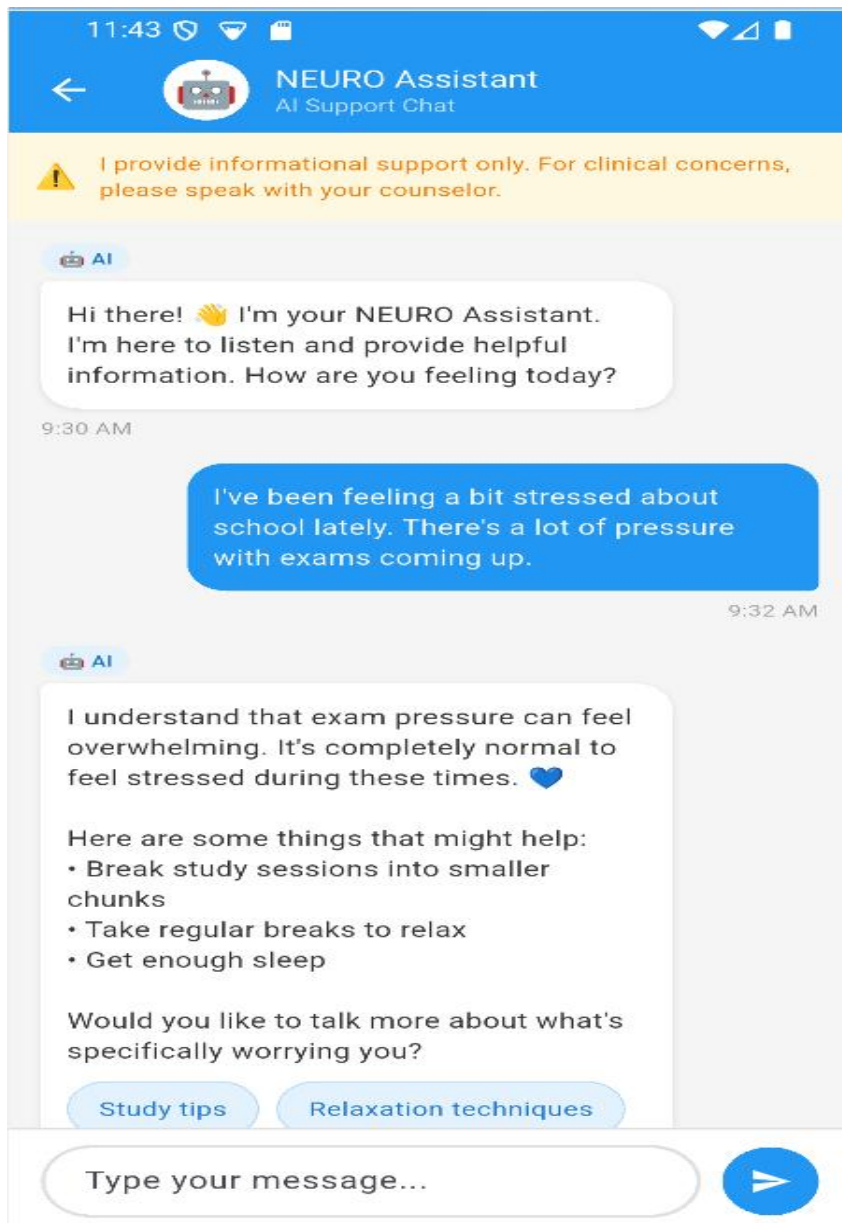


Figure 20 AI Chat Interface Screen

Description: The conversational AI support interface, displaying chat history between the adolescent and NEURO Assistant, with informational disclaimer, suggestion chips for common topics, and message input functionality.

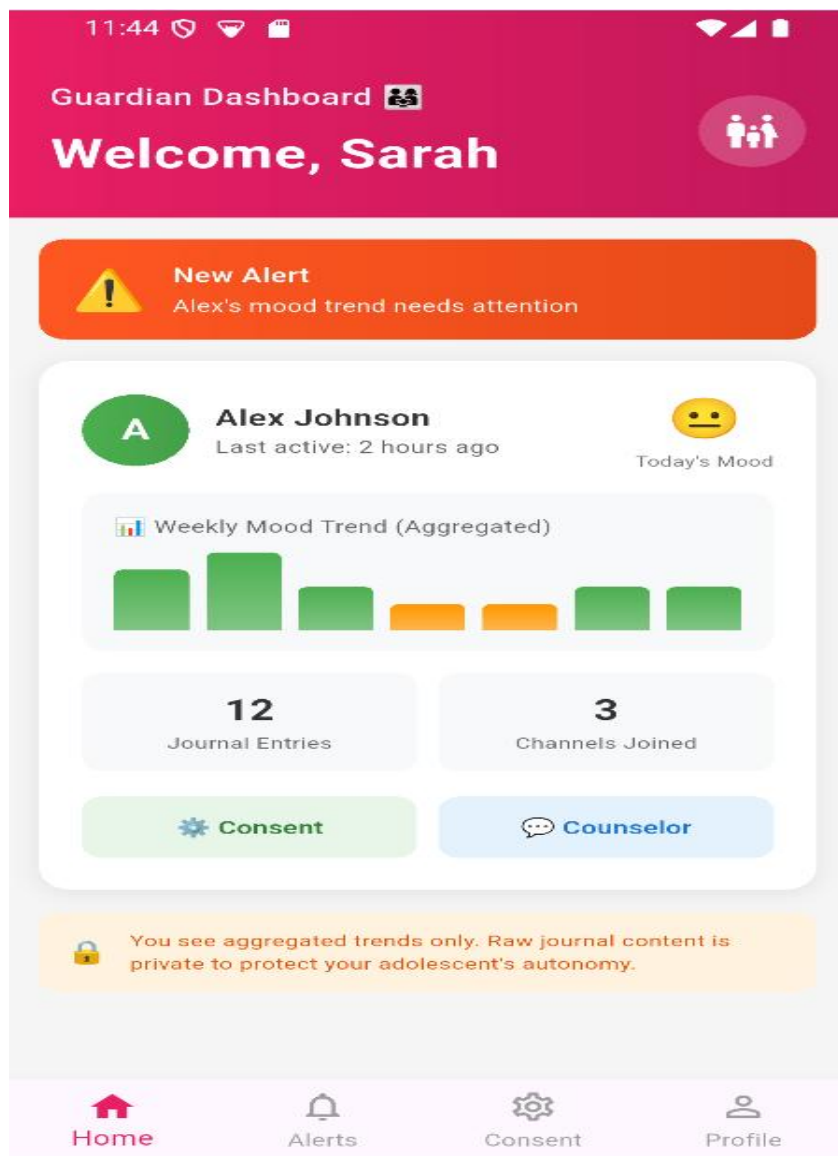


Figure 21 Guardian Dashboard Screen

Description: The monitoring dashboard for guardians, showing alert notifications, adolescent profile summary, aggregated weekly mood trends, activity statistics, and quick access to consent management and counselor communication.

WEB APPLICATION SCREENS

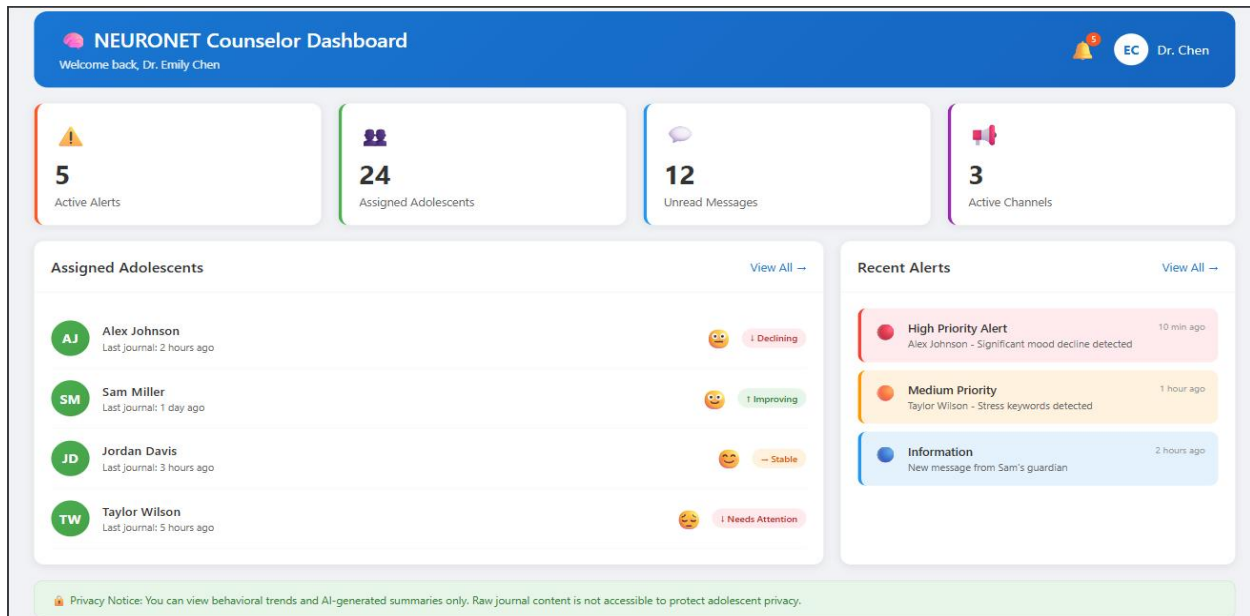


Figure 22 Counselor Dashboard Screen

Description: The professional web dashboard for counselors, featuring overview statistics (active alerts, assigned adolescents, unread messages, active channels), a detailed list of assigned adolescents with mood indicators and trend analysis, recent alert notifications with priority levels, and privacy compliance notice.

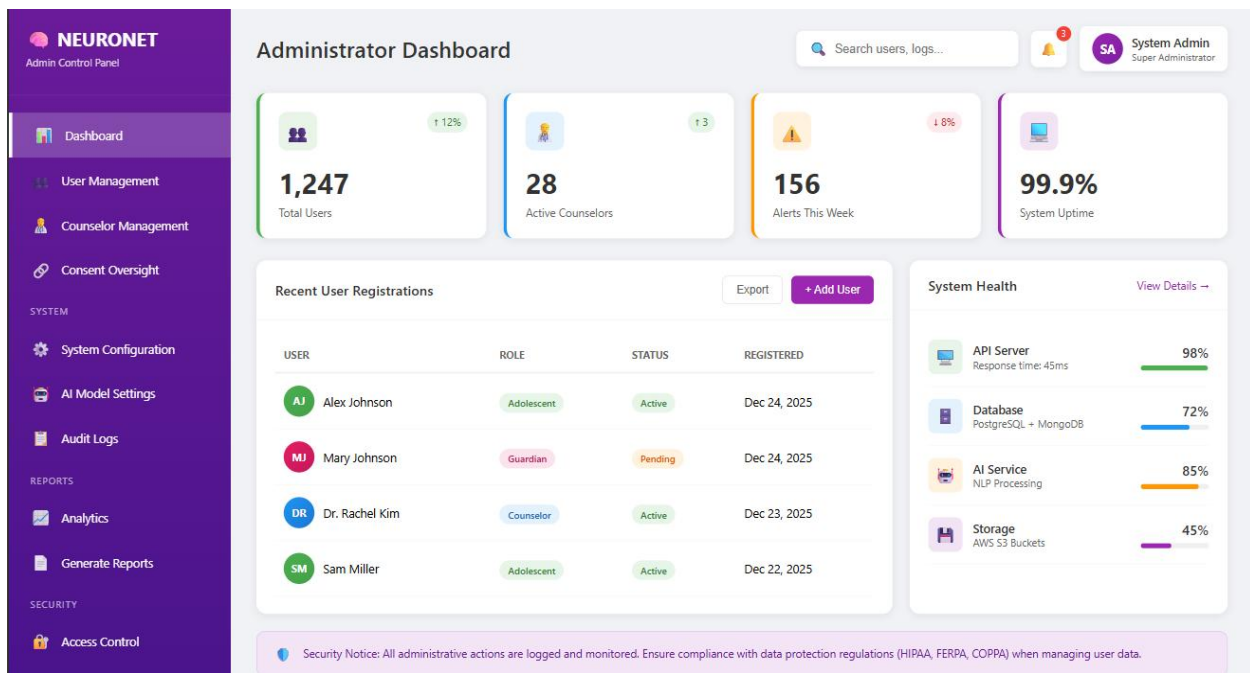


Figure 23 Administrator Dashboard Screen

Description: The comprehensive system administration interface, including sidebar navigation menu, system-wide statistics, user management table with role-based filtering, system health monitoring for API, database, AI services, and storage, audit log preview, and security compliance banner for HIPAA, FERPA, and COPPA regulations.

4.3 Screen Images and Description

4.3.1 Adolescent Mobile Interface Screens

Table 20 Adolescent Mobile Interface Screen Descriptions

Screen Name	Description	Key Elements
Login Screen	Entry point for adolescent users with a simple, welcoming design and clear branding.	Email field, password field, login button, forgot password link
Dashboard	Main hub displaying mood selector, weekly trends, and shortcuts to key features.	Mood buttons, trend chart, action cards, bottom navigation
Journal Entry	Private journaling interface with mood selection and visible privacy indicators.	Mood selector, text input area, save button, privacy notice
Mood Recording	Quick mood check-in screen using emoji-based selection.	Emoji mood options, timestamp, mood history
AI Chat	Informational AI chat interface with clear limitations notice.	Chat bubbles, input field, AI disclaimer
Personal Dashboard	Detailed view of personal trends and insights over time.	Charts, statistics, date range selector
Counselor Chat	Secure messaging interface with assigned counselor (with consent).	Message list, input field, attachment option
Channel Browser	Browse and interact with counselor-created informational channels.	Channel cards, follow buttons, content preview

4.3.2 Guardian Mobile Application Screens

Table 21 Guardian Mobile Interface Screen Descriptions

Screen Name	Description	Key Elements
Dashboard	Overview of registered adolescents with activity summaries and alerts.	Adolescent cards, alert summary, quick actions
Registration	Form for registering new adolescent accounts with consent configuration.	Input fields, consent checkboxes, submit button
Consent Management	Interface for managing data sharing and communication permissions.	Toggle switches, consent history, save button

Screen Name	Description	Key Elements
Aggregated Insights	Summarized charts showing adolescent trends without raw content access.	Trend charts, mood distribution, date filter
Alert History	List of alerts with status and actions.	Alert list, filters, mark-as-read option
Counselor Messages	Communication interface with assigned counselor.	Message thread, compose button, attachments

4.3.3 Counselor Web Dashboard Screens

Table 22 Counselor Web Dashboard Screen Descriptions

Screen Name	Description	Key Elements
Dashboard	Overview of assigned adolescents with priority alerts and trends.	Priority alerts, status overview, quick actions
Assigned Adolescents	Detailed list of assigned adolescents with trend indicators.	Sortable table, trend icons, action buttons
Adolescent Detail	Individual adolescent profile with summarized trends only.	Trend charts, consent status, session history
Alert Management	Interface for reviewing and responding to alerts.	Alert list, severity filters, response actions
Chat Interface	Secure messaging with adolescents and guardians.	Message thread, compose field, approval indicator
Channel Management	Creation and management of informational channels.	Channel list, create form, subscriber count
Channel Editor	Interface for composing and publishing channel content.	Rich text editor, publish button, preview

4.3.4 Administrator Web Dashboard Screens

Table 23 Administrator Web Dashboard Screen Descriptions

Screen Name	Description	Key Elements
Dashboard	System overview displaying metrics and health indicators.	Metric cards, system health, quick links
User Management	Interface for managing users and roles.	User table, filters, add/edit/deactivate
Assignment Manager	Tool for assigning counselors with load balancing.	Counselor list, capacity indicators, assign button
System Configuration	Interface for system parameters and thresholds.	Configuration forms, save button, defaults
Audit Logs	Searchable system activity logs for compliance.	Log table, search, date filters, export
System Monitoring	Real-time monitoring of system health and performance.	Performance charts, error logs, alerts

4.3.5 Navigation and Interaction Patterns

Mobile Navigation

- Bottom tab bar for primary navigation
- Back button for hierarchical navigation
- Pull-to-refresh functionality
- Swipe gestures for quick actions

Web Navigation

- Top navigation bar with role-specific menus
- Sidebar navigation for sections
- Breadcrumbs for deep navigation
- Keyboard shortcuts for advanced users

Common Interaction Patterns

- Confirmation dialogs for destructive actions
- Toast notifications for feedback
- Loading indicators for asynchronous actions
- Helpful empty-state guidance

CHAPTER FIVE

REQUIREMENT MATRIX

5.1 Requirement Traceability Matrix

5.1.1 Overview

The Requirement Traceability Matrix (RTM) provides a systematic mapping between the functional requirements defined in the Software Requirements Specification (SRS) and the system components designed to satisfy them. This matrix ensures complete coverage of all requirements and supports verification, validation, and impact analysis throughout the system lifecycle.

5.1.2 Functional Requirement Traceability

User Registration and Authentication

Table 24 Functional Requirement Traceability Matrix

Req ID	Requirement Description (Summary)	Source (SRS Section)	Verification Method
FR-01	Guardian-initiated adolescent registration	3.1.1 Registration & Authentication	Registration workflow test
FR-02	Guardian provides valid identification and contact info	3.1.1 Registration & Authentication	Input validation test
FR-03	Adolescent account activation via password	3.1.1 Registration & Authentication	Account activation test
FR-04	Counselor registration with admin approval	3.1.1 Registration & Authentication	Approval workflow test
FR-05	User authentication via username/password	3.1.1 Registration & Authentication	Login test
FR-06	Role-based authentication enforcement	3.1.1 Registration & Authentication	Access control test

Req ID	Requirement Description (Summary)	Source (SRS Section)	Verification Method
FR-07	Access denial on authentication failure	3.1.1 Registration & Authentication	Invalid login test
FR-08	Secure logout functionality	3.1.1 Registration & Authentication	Session termination test

User Profile Management

Req ID	Requirement Description	Source	Verification Method
FR-09	View profile information	3.1.2 User Profile Management	Profile retrieval test
FR-10	Update editable profile fields	3.1.2 User Profile Management	Update operation test
FR-11	Restrict modification of sensitive identity data	3.1.2 User Profile Management	Authorization test
FR-12	Secure storage of profile data	3.1.2 User Profile Management	Database security review
FR-13	Counselors manage educational pages	3.1.2 User Profile Management	Page creation test
FR-14	Adolescents follow counselor pages	3.1.2 User Profile Management	Follow/unfollow test
FR-15	Guardians view followed pages	3.1.2 User Profile Management	Visibility control test

Consent Management

Req ID	Requirement Description	Source	Verification Method
FR-16	Guardian consent required for data analysis	3.1.3 Consent Management	Consent enforcement test
FR-17	Grant/revoke counselor access to summaries	3.1.3 Consent Management	Consent toggle test
FR-18	Validate consent before data access	3.1.3 Consent Management	Access validation test
FR-19	Adolescents view consent status	3.1.3 Consent Management	Consent display test
FR-20	Record consent history	3.1.3 Consent Management	Audit log verification

Journaling and Mood Reflection

Req ID	Requirement Description	Source	Verification Method
FR-21	Create journal entries	3.1.4 Journaling	Entry creation test
FR-22	Record mood reflections	3.1.4 Journaling	Mood logging test
FR-23	View journal history and trends	3.1.4 Journaling	Data retrieval test
FR-24	Secure storage of journal/mood data	3.1.4 Journaling	Database integrity test
FR-25	Restrict access to raw journal data	3.1.4 Journaling	Access restriction test

AI-Based Interaction and Analysis

Req ID	Requirement Description	Source	Verification Method
FR-26	AI chat interface availability	3.1.5 AI Interaction	Interface availability test
FR-27	AI responds to informational queries	3.1.5 AI Interaction	Response accuracy test
FR-28	Emotionally neutral AI responses	3.1.5 AI Interaction	Response tone review
FR-29	No medical or clinical advice	3.1.5 AI Interaction	Content restriction test
FR-30	Redirect clinical questions to counselors	3.1.5 AI Interaction	Escalation test
FR-31	Recommend counselor pages	3.1.5 AI Interaction	Recommendation logic test
FR-32	Generate sentiment scores and trends	3.1.5 AI Interaction	Analysis output validation
FR-33	Present non-diagnostic indicators only	3.1.5 AI Interaction	Output review

Counselor Communication

Req ID	Requirement Description	Source	Verification Method
FR-34	Counselor communication after approval	3.1.6 Communication	Permission check test
FR-35	Assign counselors via system rules	3.1.6 Communication	Assignment logic test
FR-36	Limit counselor workload	3.1.6 Communication	Constraint validation
FR-37	Counselor-guardian communication	3.1.6 Communication	Content access test
FR-38	Secure message storage	3.1.6 Communication	Data

			security review
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Counselor Channels

Req ID	Requirement Description	Source	Verification Method
FR-39	Counselor channel creation	3.1.7 Channels	Channel creation test
FR-40	Adolescents join approved channels	3.1.7 Channels	Membership validation
FR-41	React and comment on posts	3.1.7 Channels	Interaction test
FR-42	Restrict sensitive disclosures	3.1.7 Channels	Content moderation test

Dashboards and Visualization

Req ID	Requirement Description	Source	Verification Method
FR-43	Adolescent emotional dashboard	3.1.8 Dashboards	UI visualization test
FR-44	Guardian aggregated dashboard	3.1.8 Dashboards	Consent-based access test
FR-45	Counselor dashboard for assigned users	3.1.8 Dashboards	Role-based data test
FR-46	Restrict sensitive data access	3.1.8 Dashboards	Access control validation
FR-47	Use clear visual elements	3.1.8 Dashboards	UI inspection

Alerts and Notifications

Req ID	Requirement Description	Source	Verification Method
FR-48	Generate alerts on thresholds	3.1.9 Alerts	Trigger condition test
FR-49	Notify authorized users	3.1.9 Alerts	Notification routing test
FR-50	View alert history	3.1.9 Alerts	Log retrieval test
FR-51	Avoid diagnostic implications	3.1.9 Alerts	Content review

Administrative Management

Req ID	Requirement Description	Source	Verification Method
FR-52	Admin manage users and roles	3.1.10 Administration	Role management test
FR-53	Admin assign counselors	3.1.10 Administration	Assignment workflow test
FR-54	Monitor system logs and performance	3.1.10 Administration	Monitoring verification
FR-55	Restrict admin access to private data	3.1.10 Administration	Access restriction audit

Data Security

Req ID	Requirement Description	Source	Verification Method
FR-56	Enforce role-based access control for all system features	3.1.11 Data Security and Access Control	RBAC validation test
FR-57	Ensure sensitive data is accessible only to authorized users	3.1.11 Data Security and Access Control	Access control audit
FR-58	Log significant system actions for monitoring and auditing	3.1.11 Data Security and Access Control	Logging and audit review

5.1.3 Non-Functional Requirement Traceability

Performance

Table 25 Non-Functional Requirement Traceability Matrix

Req ID	Requirement Description	Source	Verification Method
NFR-01	Respond to user requests within acceptable response times	3.5.1 Performance Requirements	Response time measurement test
NFR-02	Support concurrent access without performance degradation	3.5.1 Performance Requirements	Load testing with concurrent users

NFR-03	AI sentiment analysis processed efficiently	3.5.1 Performance Requirements		AI processing latency test
Req ID	Requirement Description	Source	Verification Method	
NFR-04	Manage server resources effectively during peak usage	3.5.1 Performance Requirements	Stress and peak load testing	
NFR-05	Scalable to accommodate user growth	3.5.1 Performance Requirements	Scalability simulation test	
NFR-06	Respond within 2 seconds for 95% of interactive requests	3.5.1 Performance Requirements	Response time SLA validation	
NFR-07	Support 100+ concurrent active users during pilot	3.5.1 Performance Requirements	Concurrent user load test	
NFR-08	AI results available within 24 hours of submission	3.5.1 Performance Requirements	AI pipeline timing audit	
NFR-09	Alert evaluation max delay of 24 hours for trends	3.5.1 Performance Requirements	Alert processing delay test	
NFR-10	DB operations within 500ms for 95% of transactions	3.5.1 Performance Requirements	Database performance benchmark	

Usability

Req ID	Requirement Description	Source	Verification Method
NFR-11	Intuitive interface for adolescent users	3.5.2 Usability Requirements	Usability testing with target group
NFR-12	Clear, simple, and supportive language	3.5.2 Usability Requirements	Content and UI language review
NFR-13	Consistent navigation and layout across platforms	3.5.2 Usability Requirements	Cross-platform UI consistency audit
NFR-14	Minimize steps for common tasks	3.5.2 Usability Requirements	Task completion efficiency test
NFR-15	Supportive, non-stigmatizing presentation of insights/alerts	3.5.2 Usability Requirements	User feedback and perception survey
NFR-16	Operational availability of at least 99.5%	3.5.2 Usability Requirements	Uptime and availability monitoring

Security

Req ID	Requirement Description	Source	Verification Method
NFR-17	Require authentication before access	3.5.3 Security Requirements	Authentication bypass test
NFR-18	Enforce strict role-based access control	3.5.3 Security Requirements	RBAC validation test
NFR-19	Encrypt sensitive data during storage and transmission	3.5.3 Security Requirements	Encryption compliance audit
NFR-20	Restrict access to private content (journals/chats)	3.5.3 Security Requirements	Access control penetration test
NFR-21	Maintain audit logs of critical actions	3.5.3 Security Requirements	Log completeness and integrity review
NFR-22	Securely store passwords using hashing/encryption	3.5.3 Security Requirements	Password storage security audit

5.2 Component-to-Requirement Mapping

5.2.1 Subsystem Coverage Summary

Table 26 Subsystem to Requirement Coverage

Subsystem	Requirements Covered
User Management	FR-01, FR-02, FR-09, FR-10, FR-11, FR-12
Authentication	/ FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-25, FR-46, FR-55, FR-

Authorization	56, FR-57
Consent Management	FR-16, FR-17, FR-18, FR-19, FR-20
Journaling / Mood Tracking	FR-21, FR-22, FR-23, FR-24
Communication / Chat	FR-26, FR-34, FR-37, FR-38
Subsystem	Requirements Covered
AI Analysis / Insight	FR-27, FR-28, FR-29, FR-30, FR-31, FR-32, FR-33
Alert / Notification	FR-48, FR-49, FR-50, FR-51
Counselor Assignment / Channel	FR-13, FR-14, FR-35, FR-36, FR-39, FR-40, FR-41, FR-42
Dashboard / Visualization	FR-15, FR-43, FR-44, FR-45, FR-47
Administration / Monitoring	FR-52, FR-53, FR-54, FR-58

5.2.2 Verification Checklist

All 58 functional requirements from the SRS are mapped to specific system components:

- User Registration and Authentication (FR-01 to FR-08): 8 requirements
- User Profile Management (FR-09 to FR-15): 7 requirements
- Consent Management (FR-16 to FR-20): 5 requirements
- Journaling and Mood Reflection (FR-21 to FR-25): 5 requirements
- AI-Based Interaction and Analysis (FR-26 to FR-33): 8 requirements
- Counselor Communication (FR-34 to FR-38): 5 requirements
- Counselor Channels (FR-39 to FR-42): 4 requirements
- Dashboards and Visualization (FR-43 to FR-47): 5 requirements
- Alerts and Notifications (FR-48 to FR-51): 4 requirements
- Administrative Management (FR-52 to FR-55): 4 requirements
- Data Security (FR-56 to FR-58): 3 requirements

Total: 58 functional requirements fully traced to design components.

5.2.3 Design Completeness Statement

This Software Design Document provides complete coverage of all functional and non-functional requirements specified in the NEURONET Software Requirements Specification. Every requirement has been traced to one or more design components, ensuring that the implemented system will satisfy all specified functionality.

The design strictly adheres to:

- Non-clinical system constraints
- Privacy and consent requirements
- Ethical AI usage guidelines
- Role-based access control principles
- Data security and encryption standards

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APPENDIX

A. Glossary of Technical Terms

- **API (Application Programming Interface):** A set of protocols and tools for building software applications that specify how software components should interact.
- **Asynchronous Processing:** A method of parallel processing where tasks are executed independently without waiting for other tasks to complete.
- **CRUD Operations:** Create, Read, Update, Delete – the four basic operations of persistent storage.
- **JWT (JSON Web Token):** A compact, URL-safe means of representing claims to be transferred between two parties for authentication.
- **Microservices:** An architectural style that structures an application as a collection of loosely coupled services.
- **ORM (Object-Relational Mapping):** A programming technique for converting data between incompatible type systems in object-oriented programming languages.
- **REST (Representational State Transfer):** An architectural style for designing networked applications using stateless communication protocols.
- **Sentiment Analysis:** The use of natural language processing to identify and extract subjective information from text.
- **TLS (Transport Layer Security):** A cryptographic protocol designed to provide communications security over a computer network.

- **UUID (Universally Unique Identifier):** A 128-bit number used to identify information in computer systems.

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